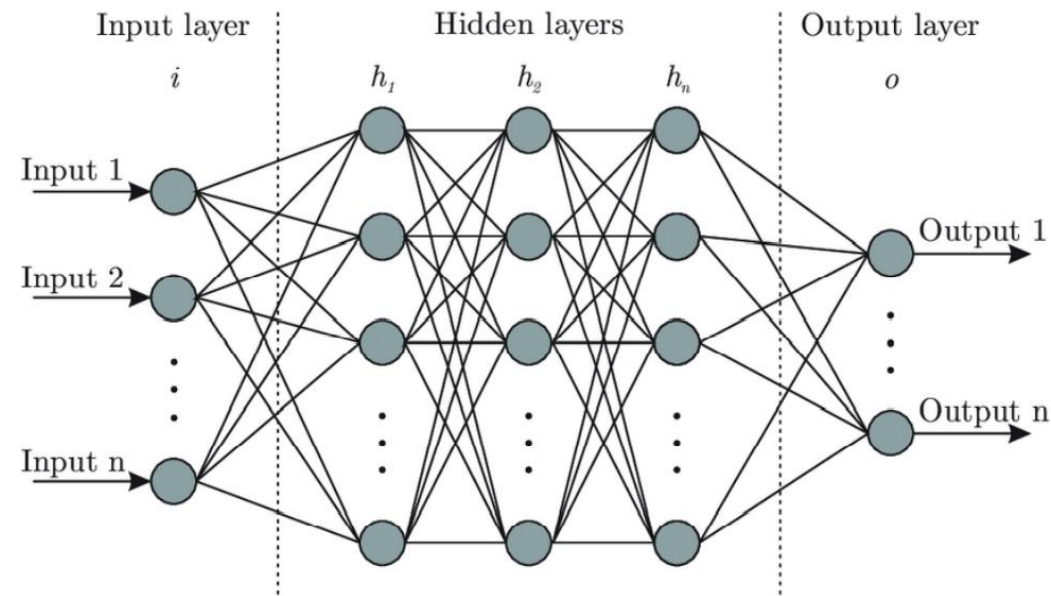




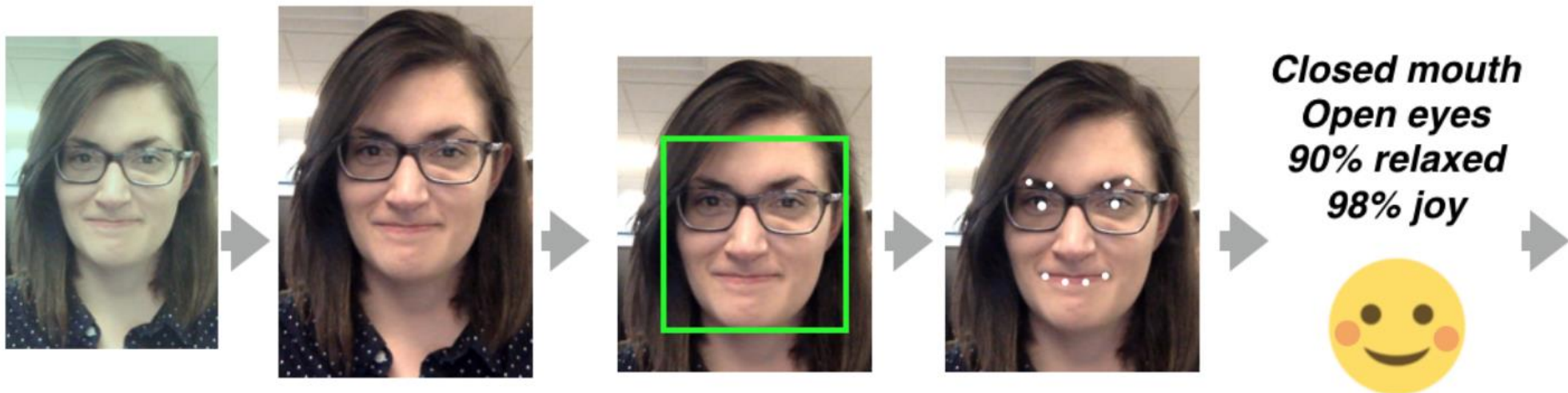
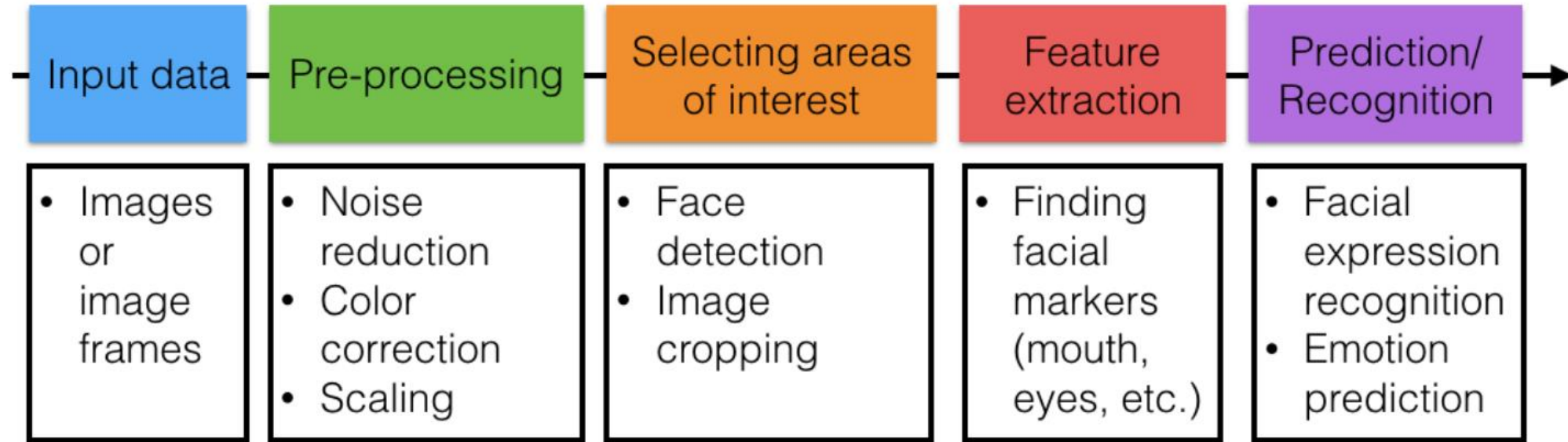
Computer vision

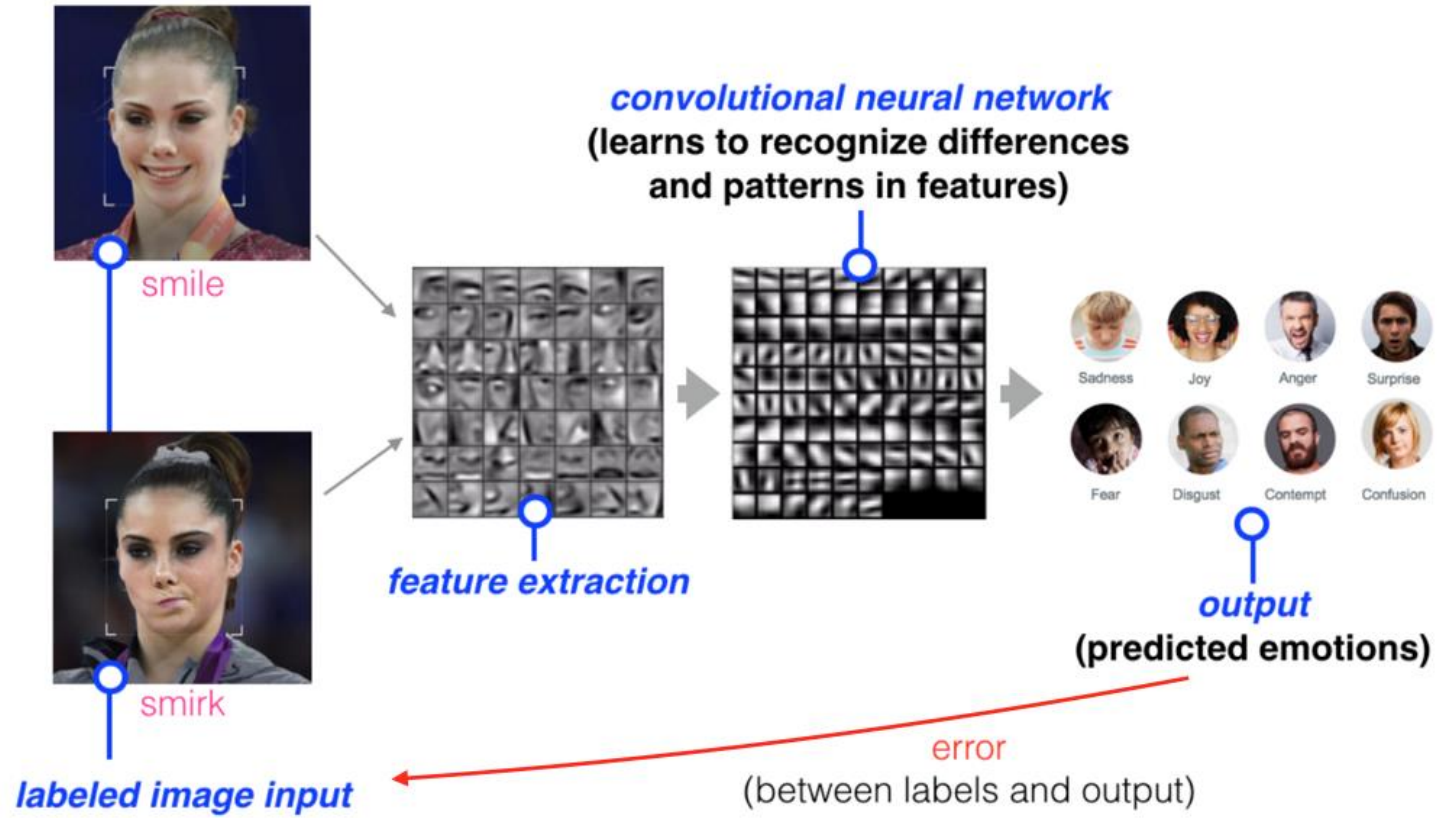
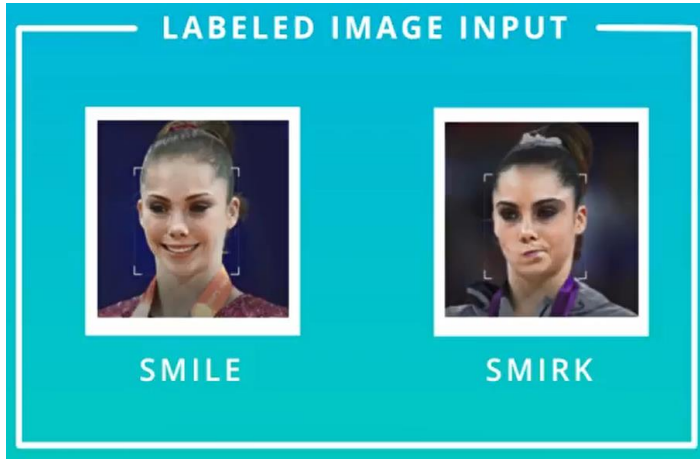
Drawbacks of MLP

- There are too many weights to train
- Lose local information and all 2D information by flattening out the image
- Will only work on very similar, well centered images

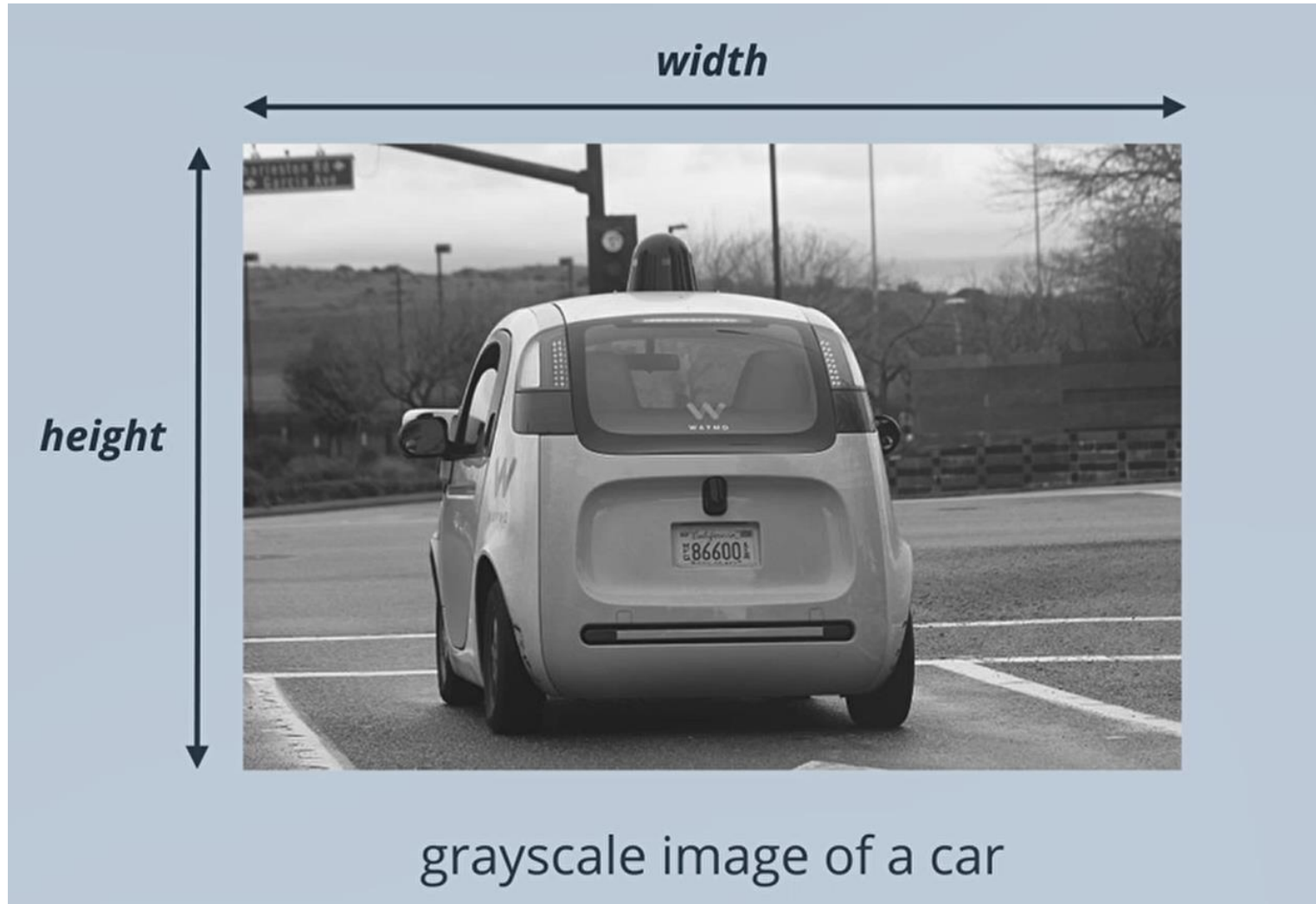


Computer vision pipeline





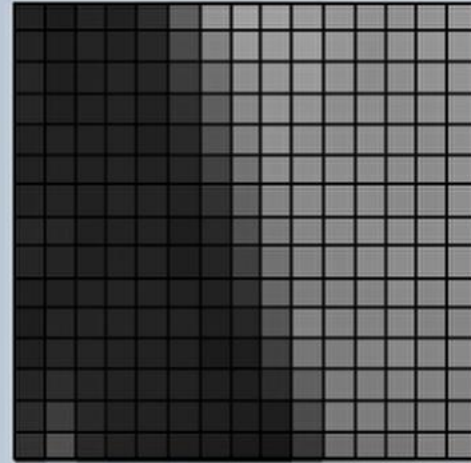
Images as Grids of Pixels

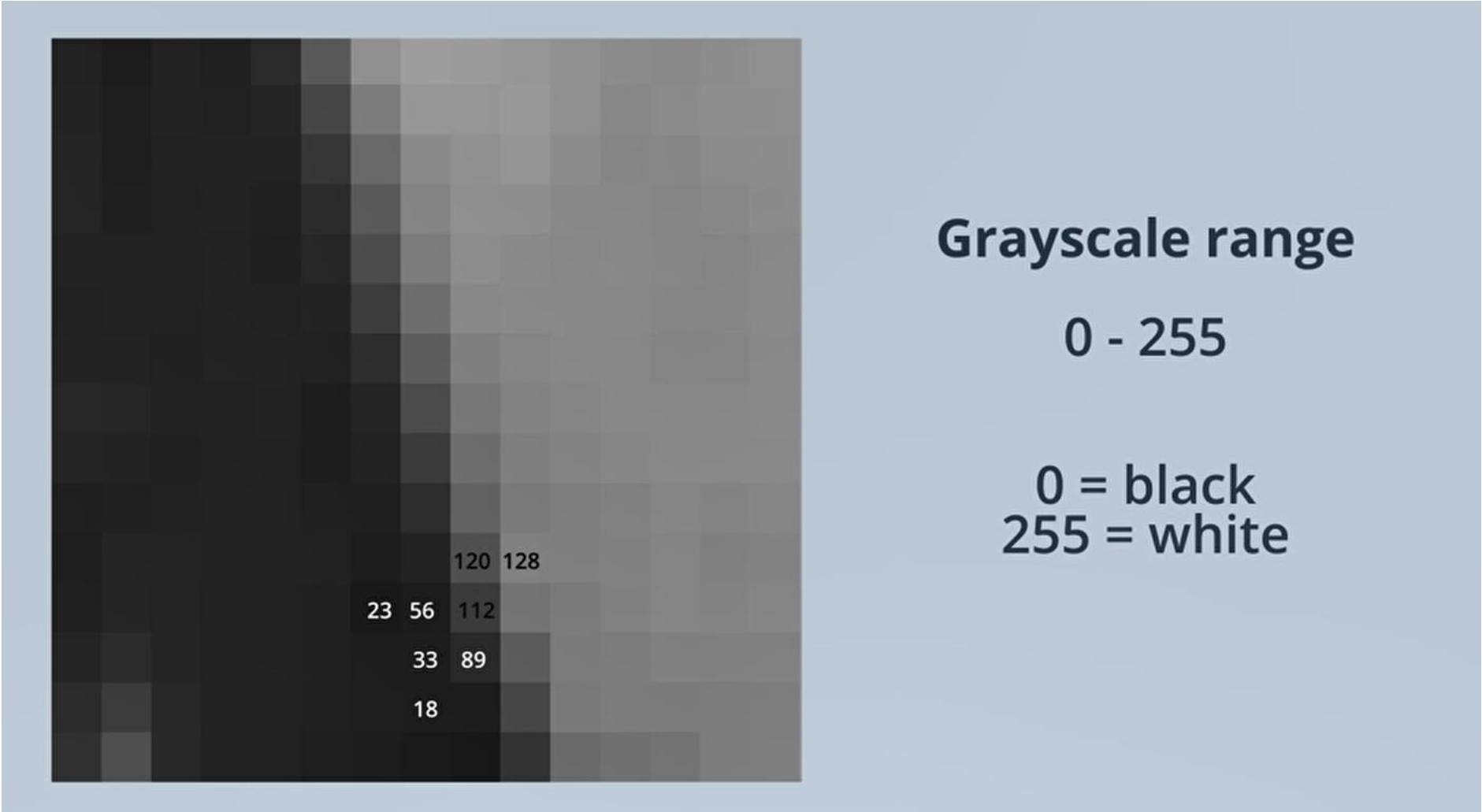


Images as Grids of Pixels



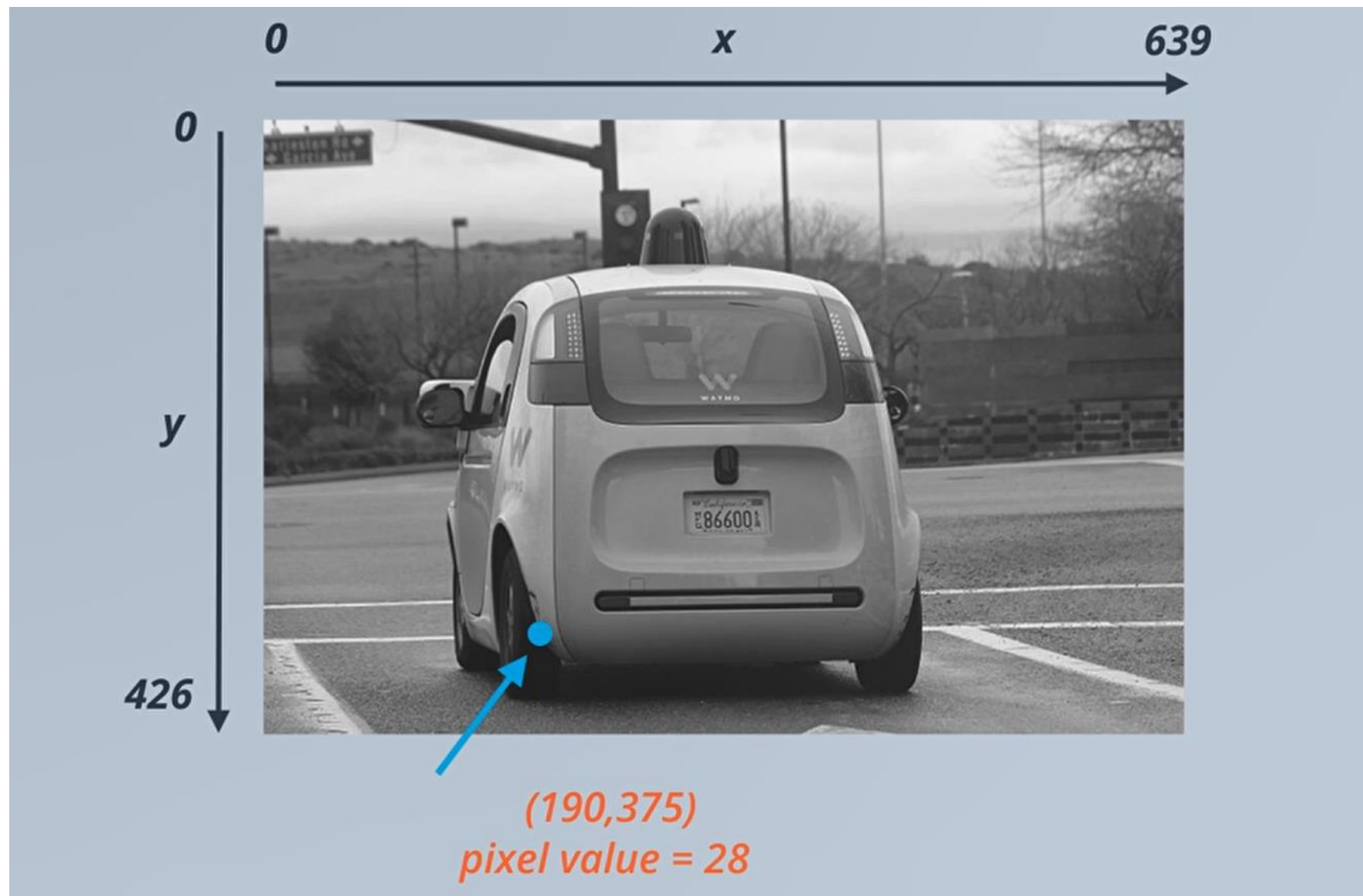
Images as Grids of Pixels



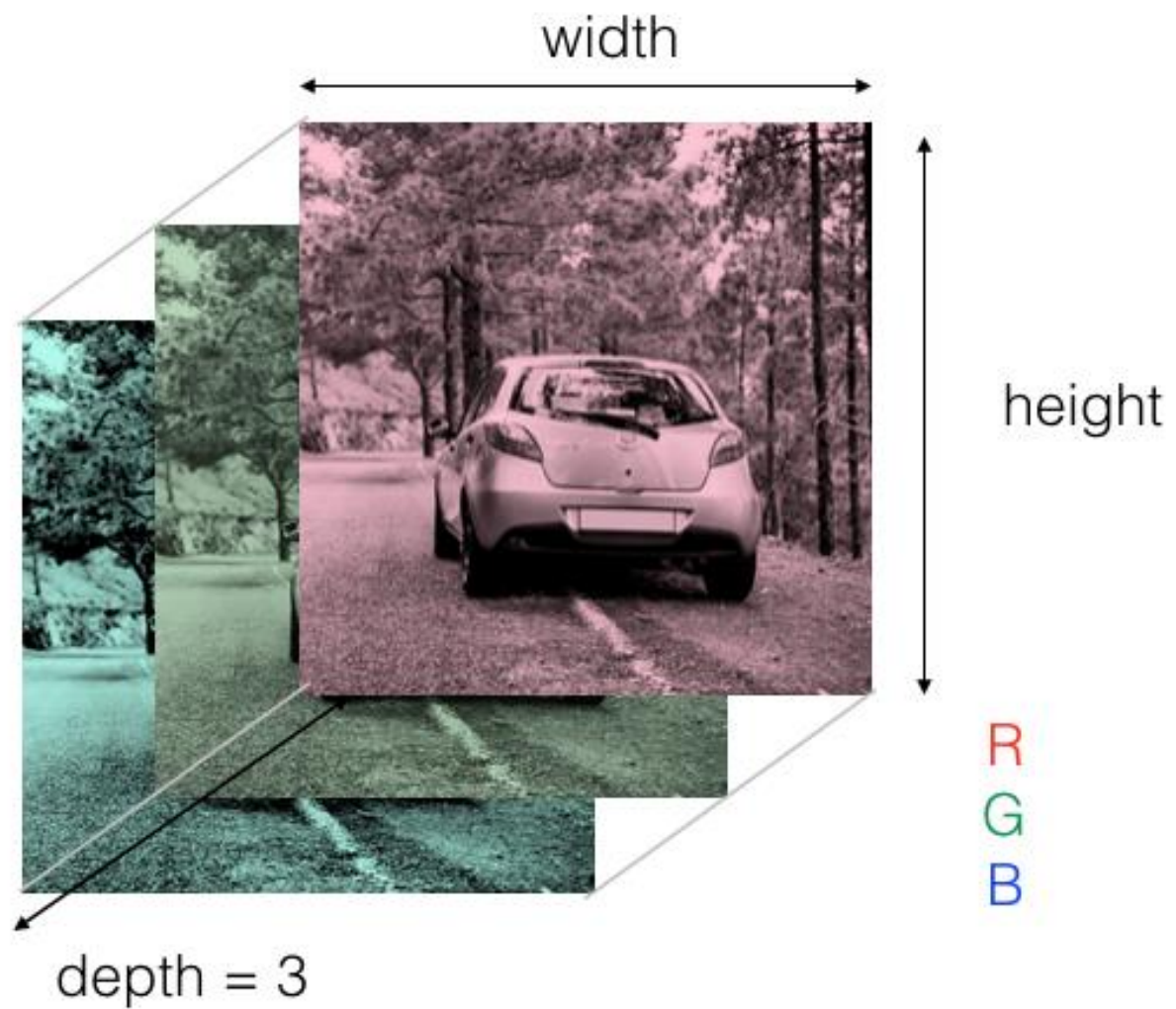


0 - 255

0 = black
255 = white



Color image



Importance of color

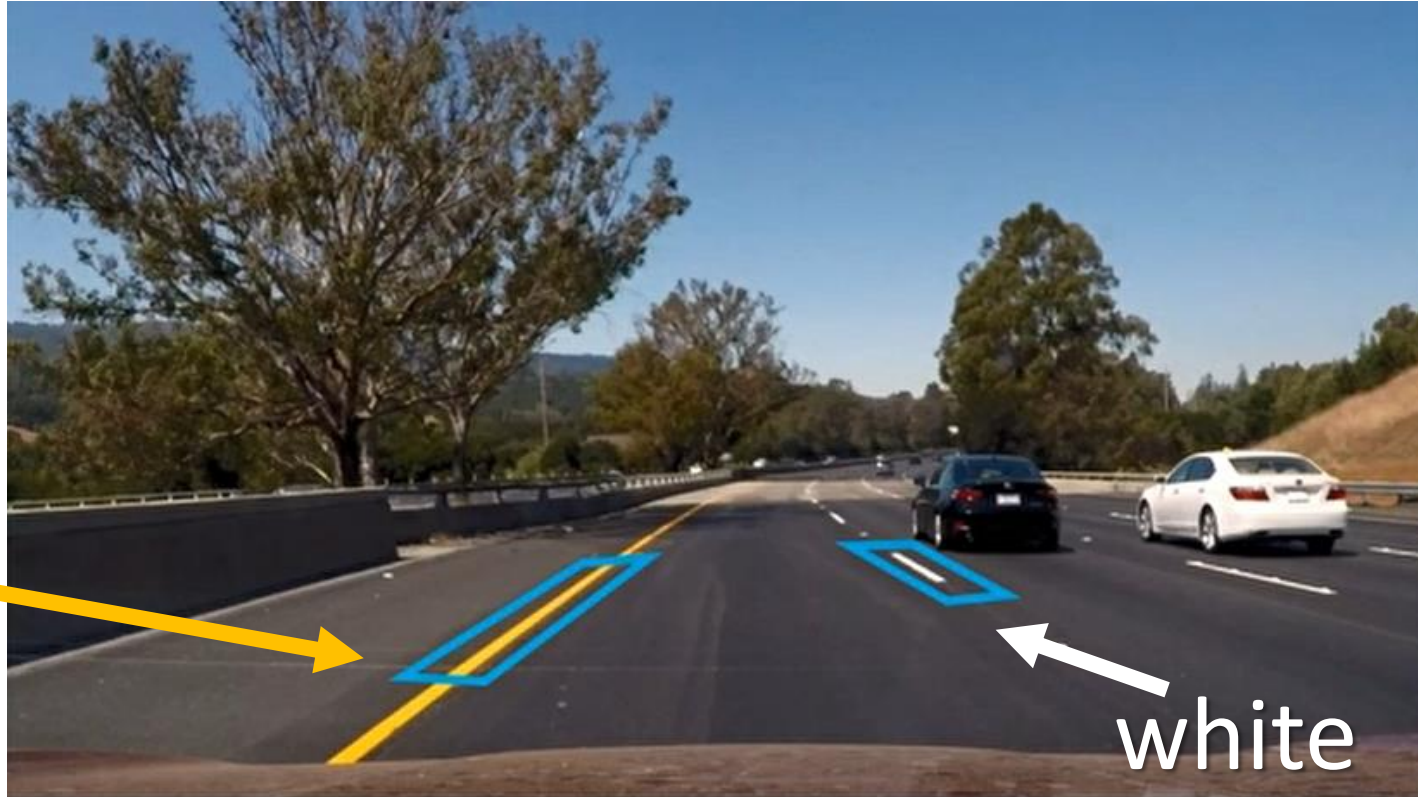


Which lane line is yellow?

Which is white?

Importance of color

Yellow



white

Color Threshold





CGI in Hollywood



CGI in Hollywood



CGI in Hollywood



CGI in Hollywood



Oops!!!



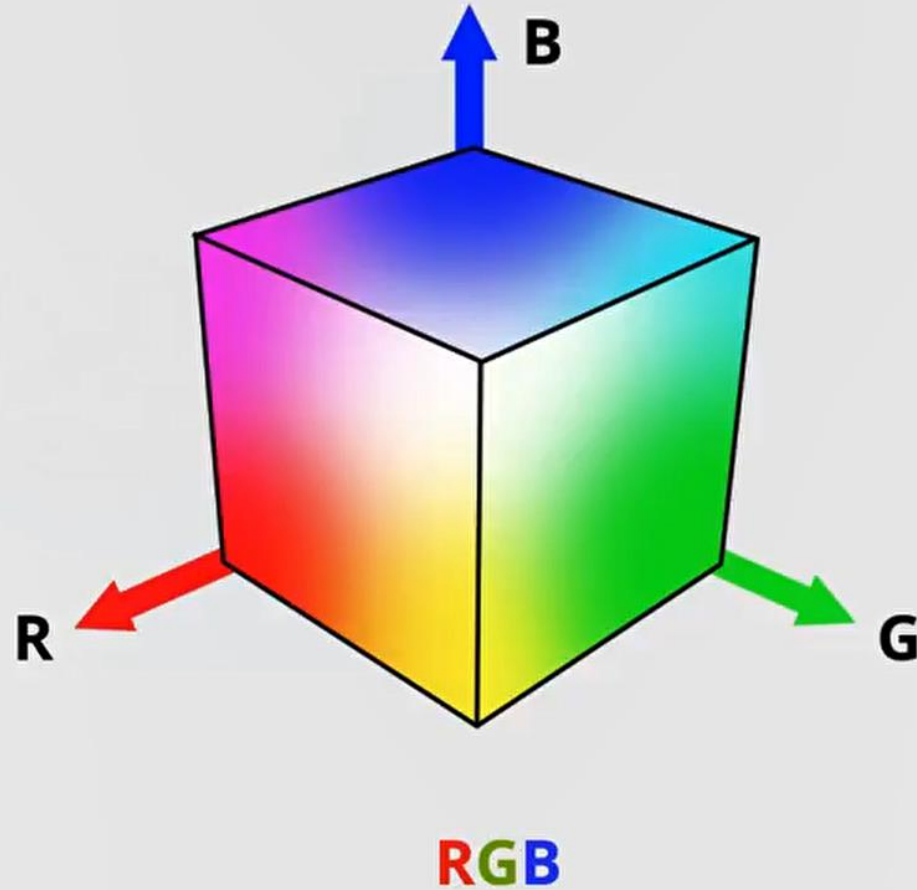
Coding Session

- Blue Screen Code

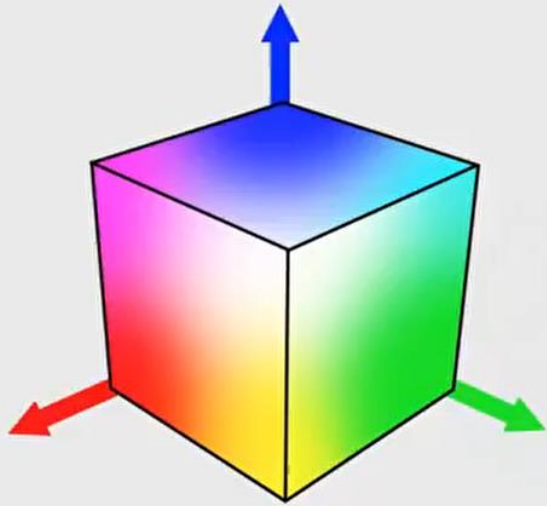




Color space



Color space



RGB

Red, Green, Blue



HSV

Hue, Saturation, Value

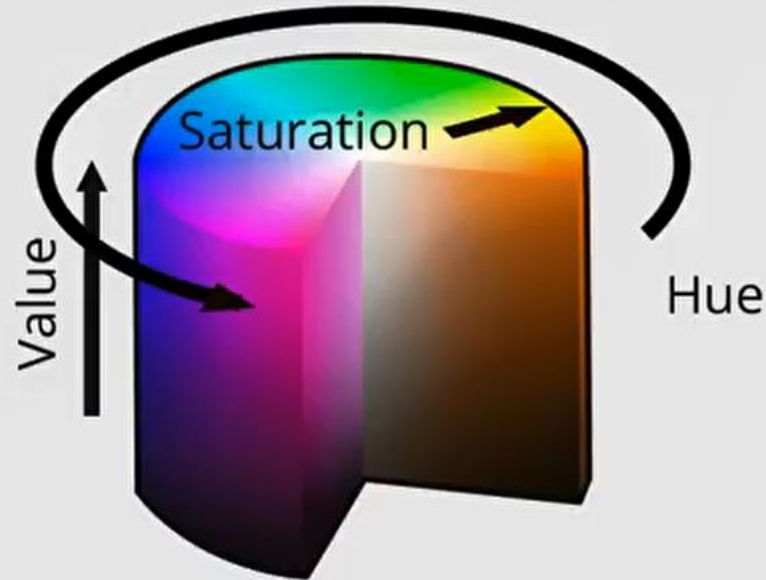


HLS

Hue, Lightness, Saturation

HSV color space

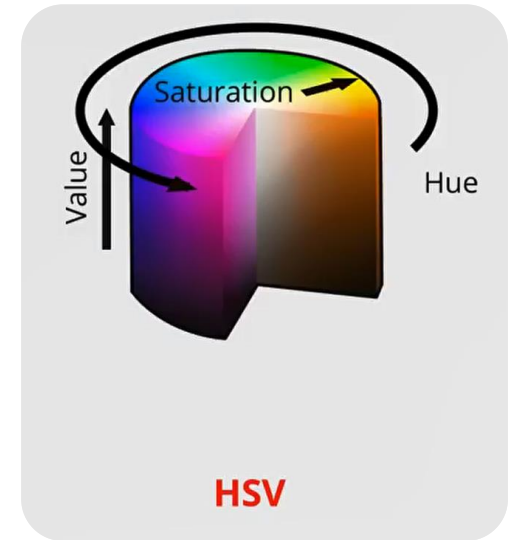
- Isolates the Value (V) component, which varies the most under different lighting conditions



- The H channel stays fairly consistent in shadow or excessive brightness

HSV

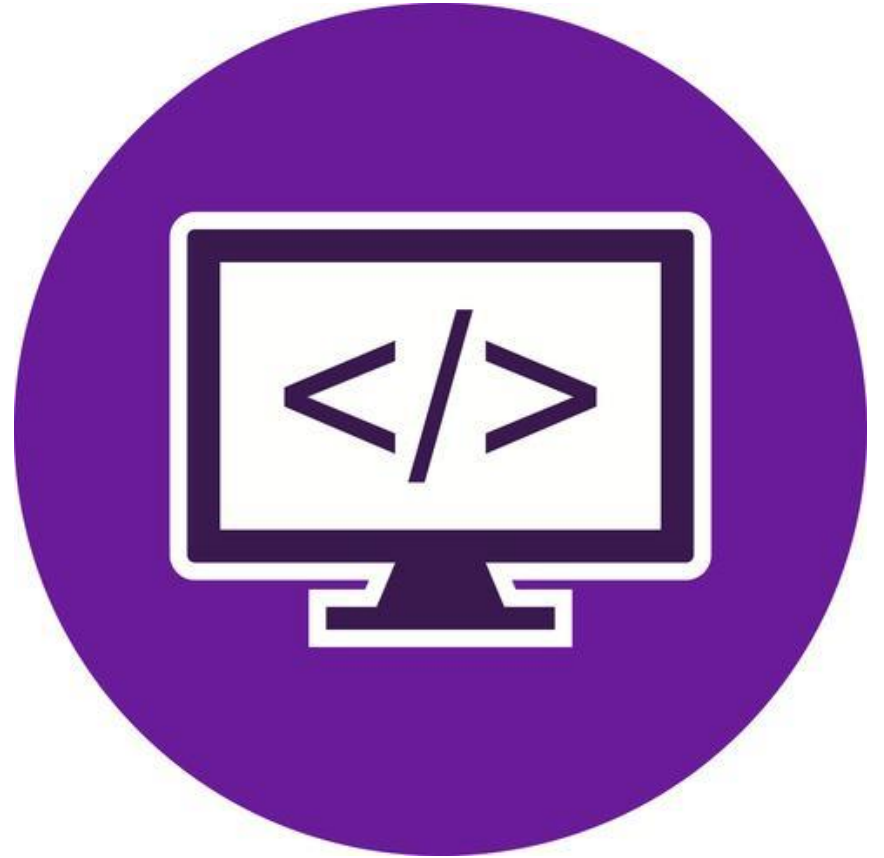
HSV color space



- The HSV colour space also has 3 channels: **Hue**, **Saturation** and **Value or intensity**.
- Hue channel represents the "color". For example, "red" would be a color.
 - Light red/dark red would not be a color.
- Saturation channel is the "amount" of color
 - (this differentiates between a pale green and pure green).
- Value or intensity is the brightness of the color
 - (light green or dark green).

Coding Session

- Color space (RGB vs HSV)



Day and Night Classification



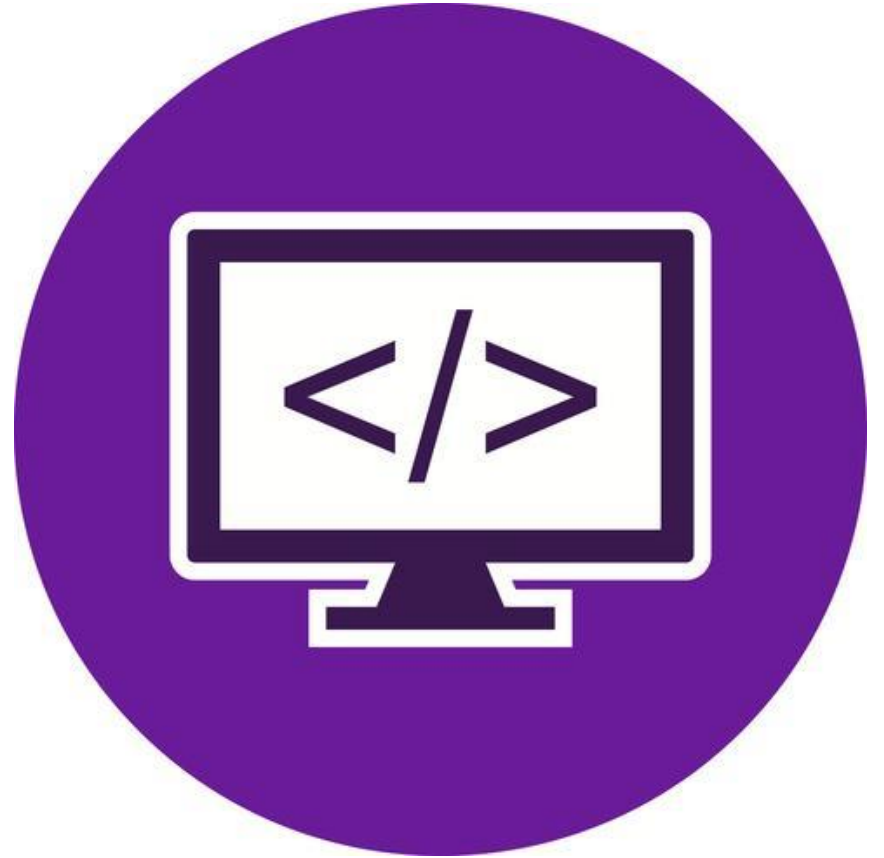
day



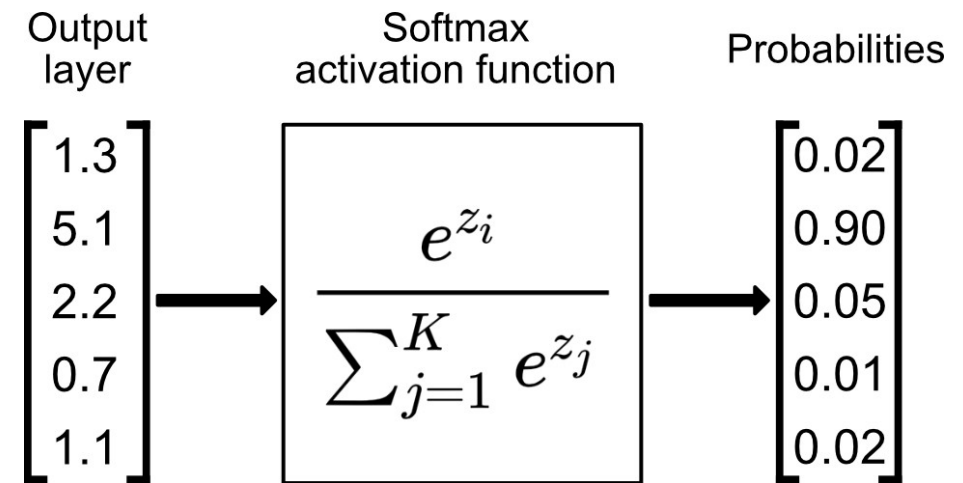
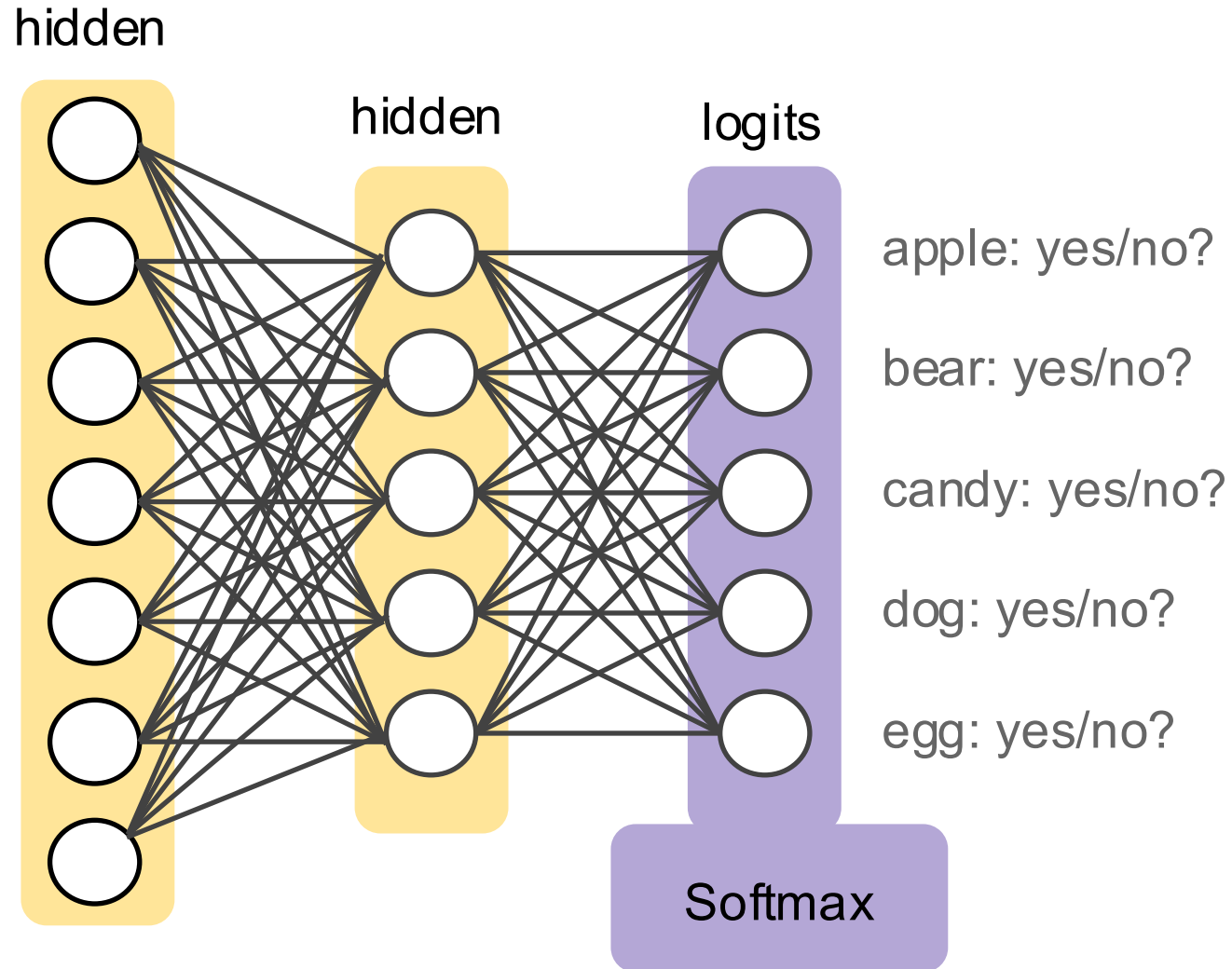
night

Coding Session

- Day and Night Classification code



Cross-Entropy vs NLL loss



Cross-Entropy vs NLL loss

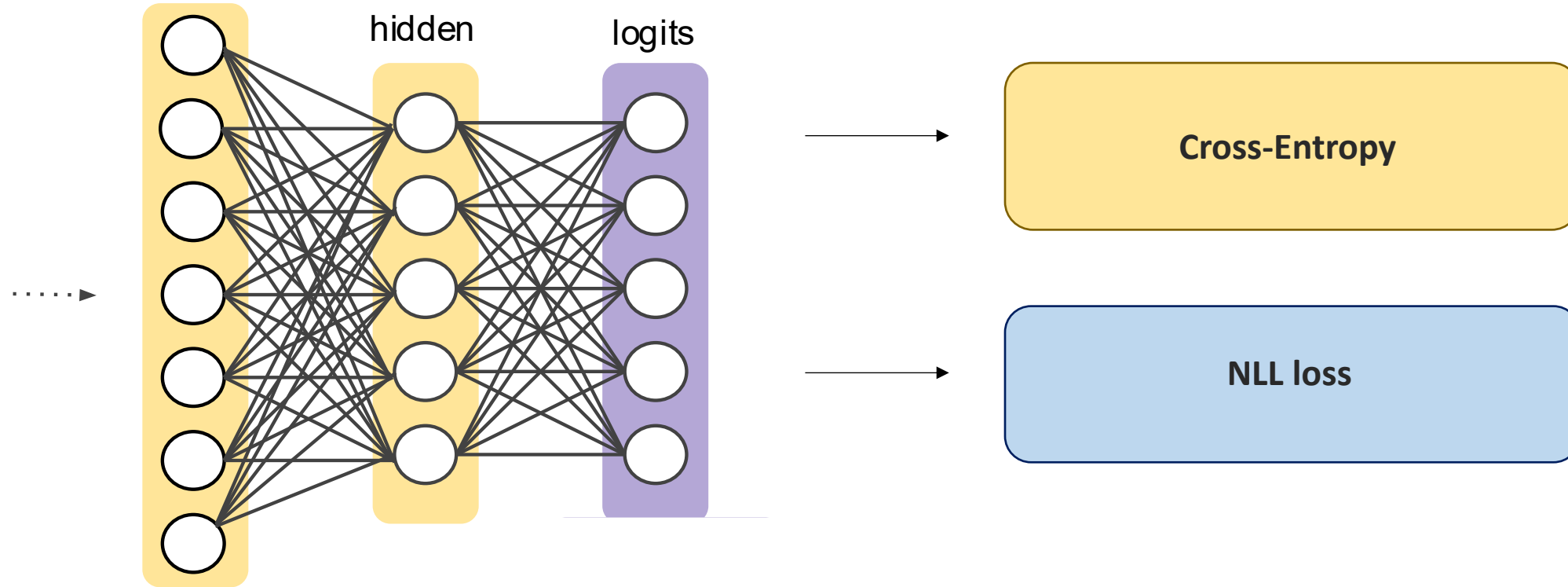
hidden

hidden

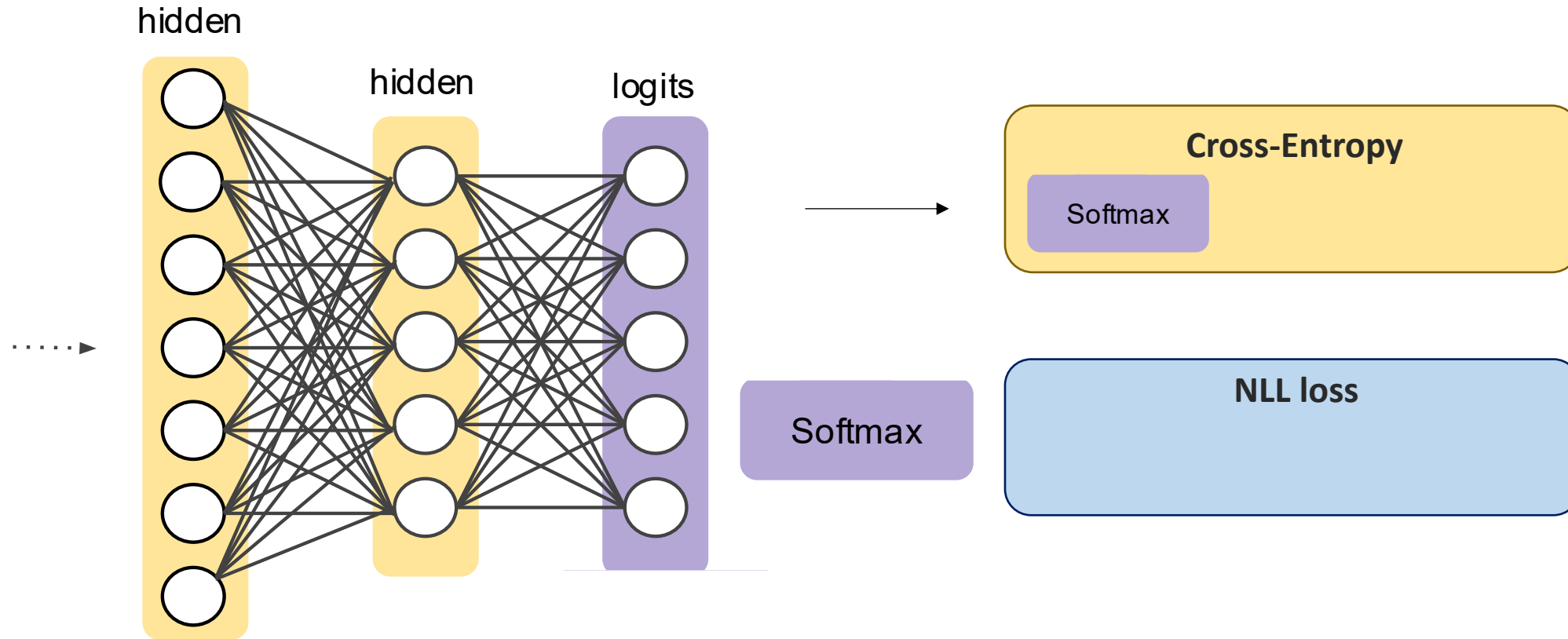
logits

Cross-Entropy

NLL loss



Cross-Entropy vs NLL loss



Types of Features



Color (Texture, Blobs)

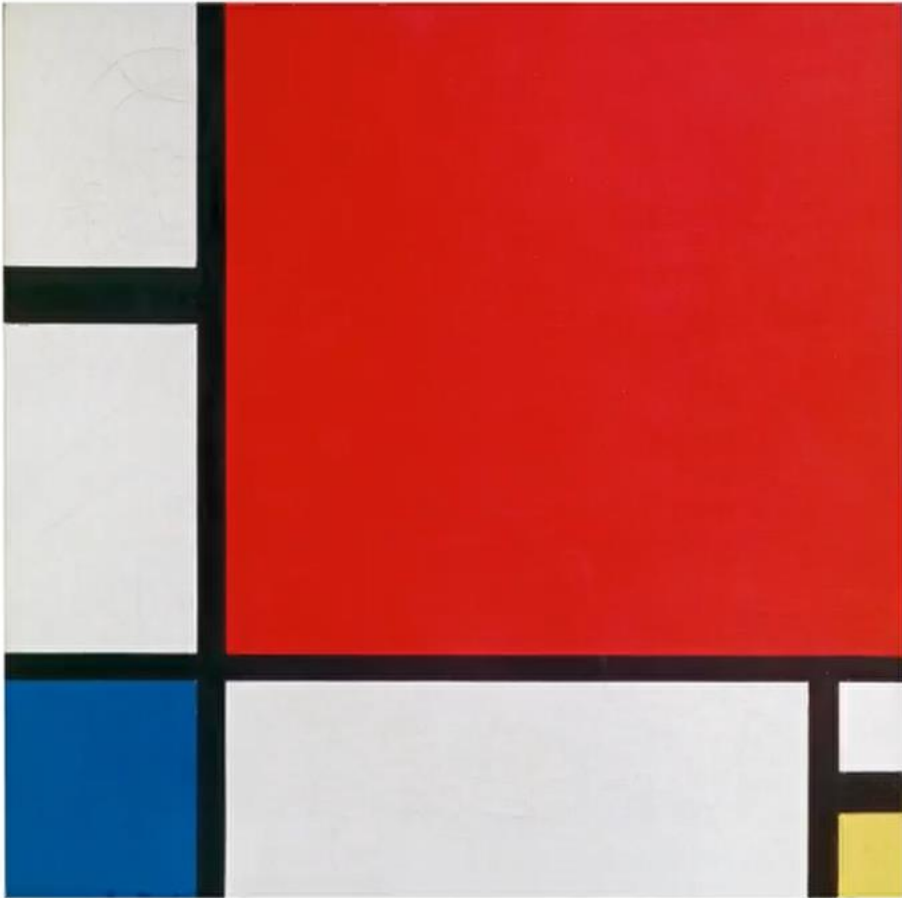
Edges

- Area with a high intensity gradient

Corners

- At the intersection of 2 Edges

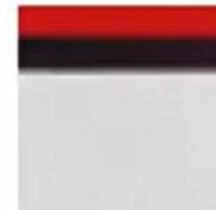
Types of Features



PATCHES



A



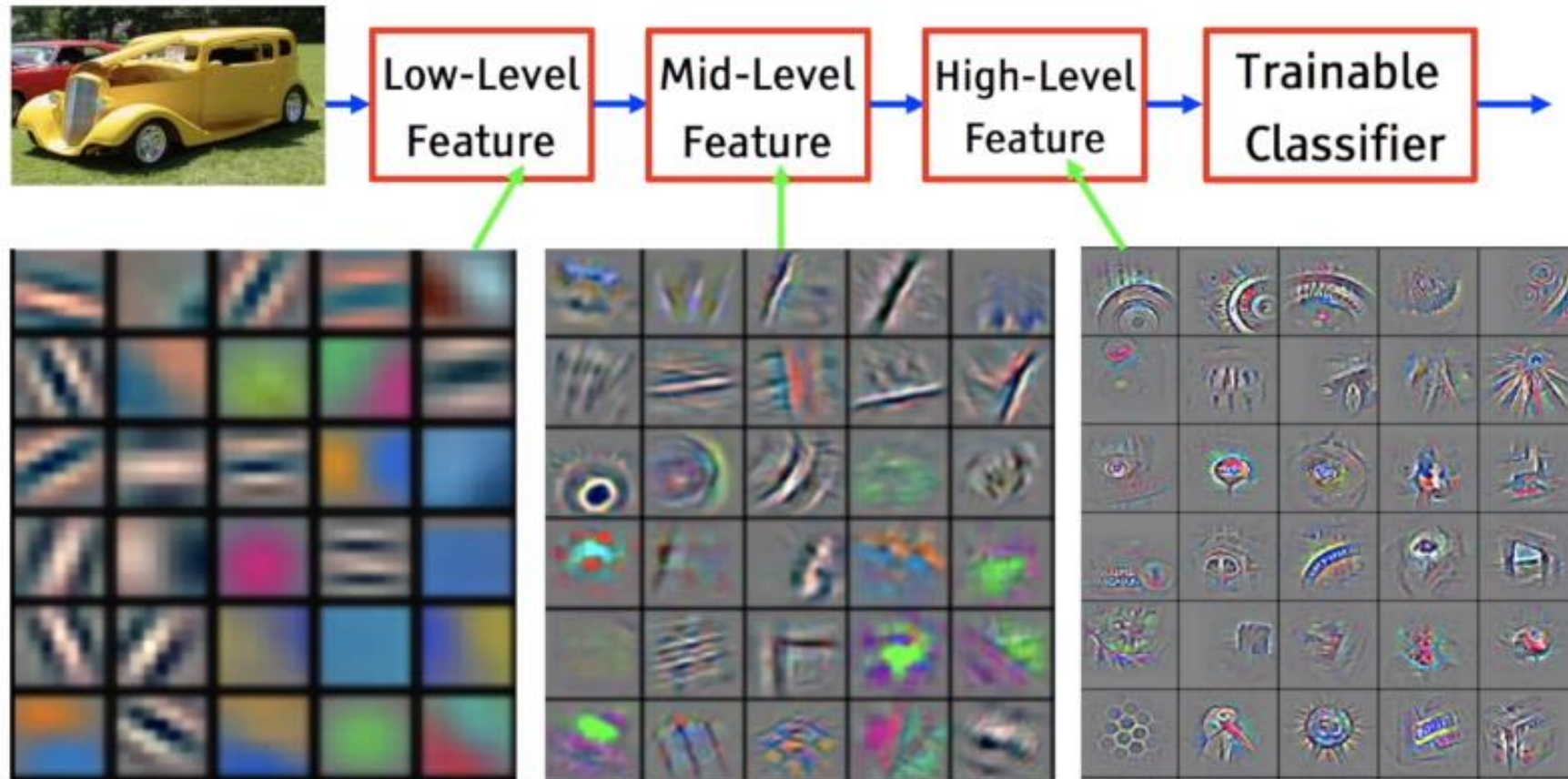
B



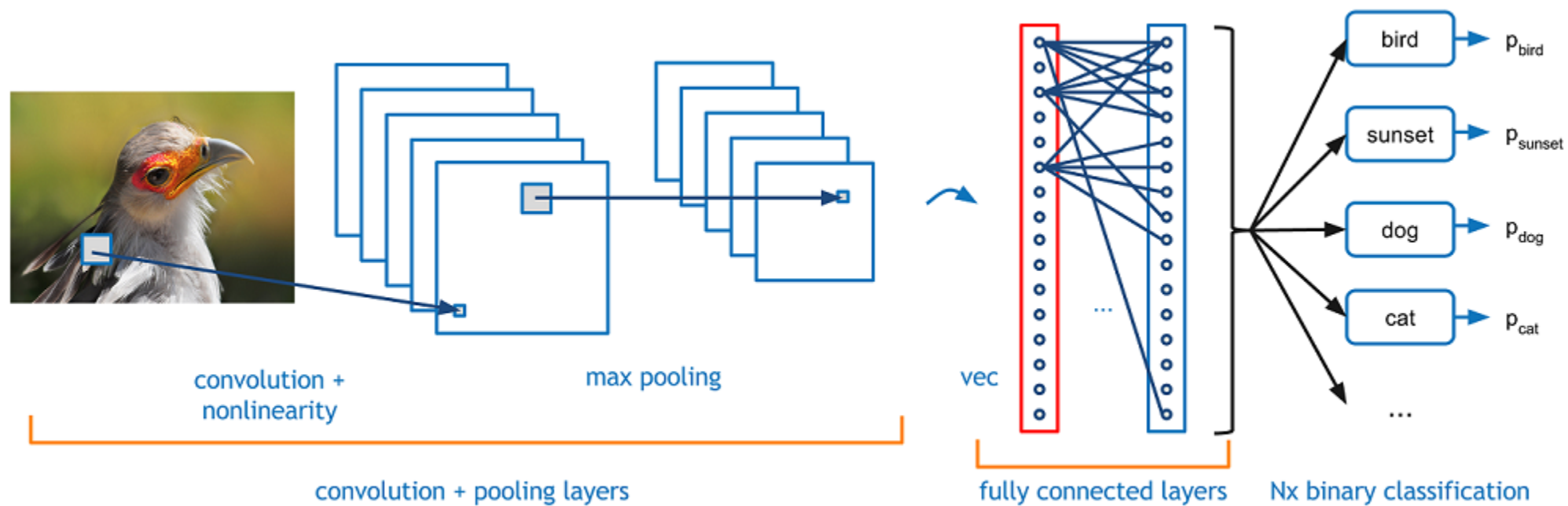
C

What area do they match with?

Learned Patterns

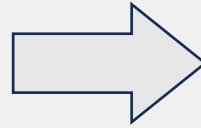


Structure



Filter

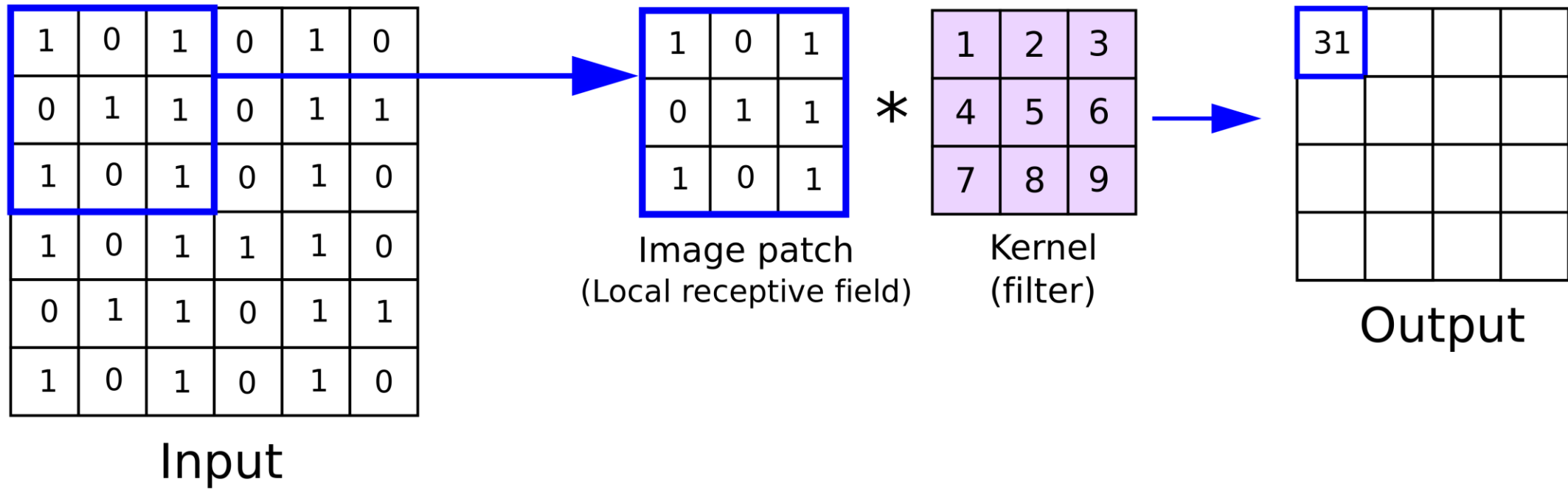
original



filtered (Gaussian filter)



Filter



Filter

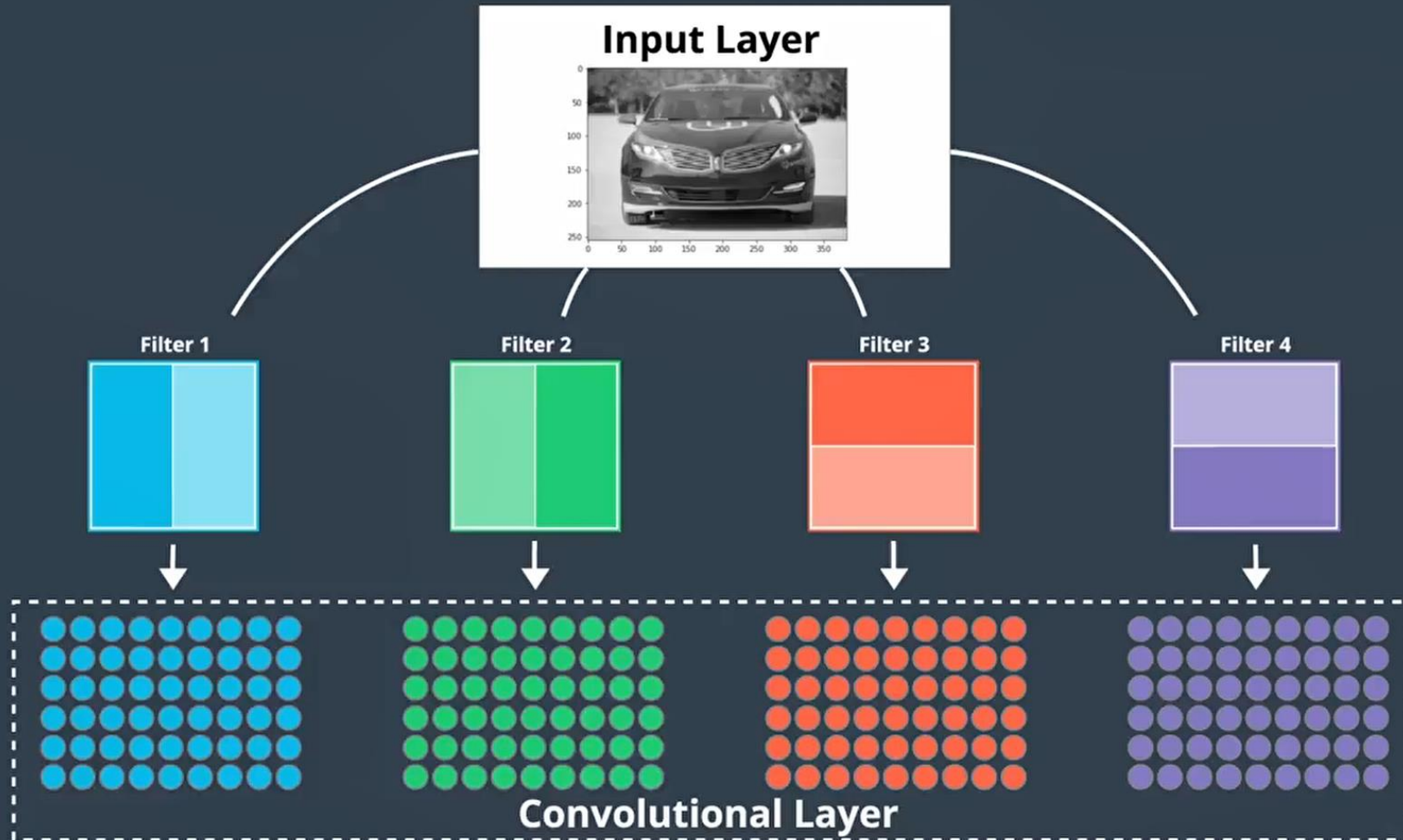
1 _{x1}	1 _{x0}	1 _{x1}	0	0
0 _{x0}	1 _{x1}	1 _{x0}	1	0
0 _{x1}	0 _{x0}	1 _{x1}	1	1
0	0	1	1	0
0	1	1	0	0

Image

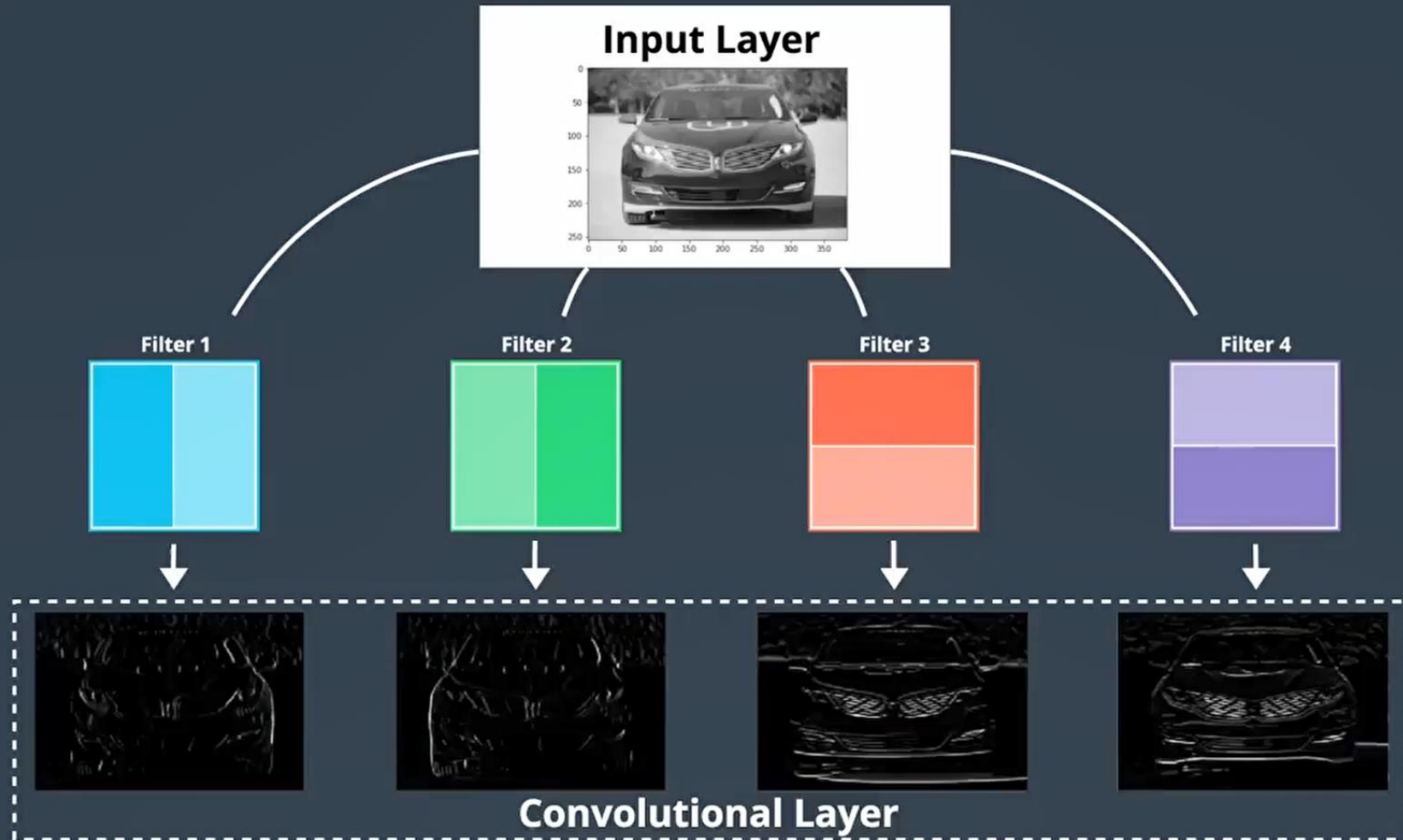
4		

Convolved
Feature

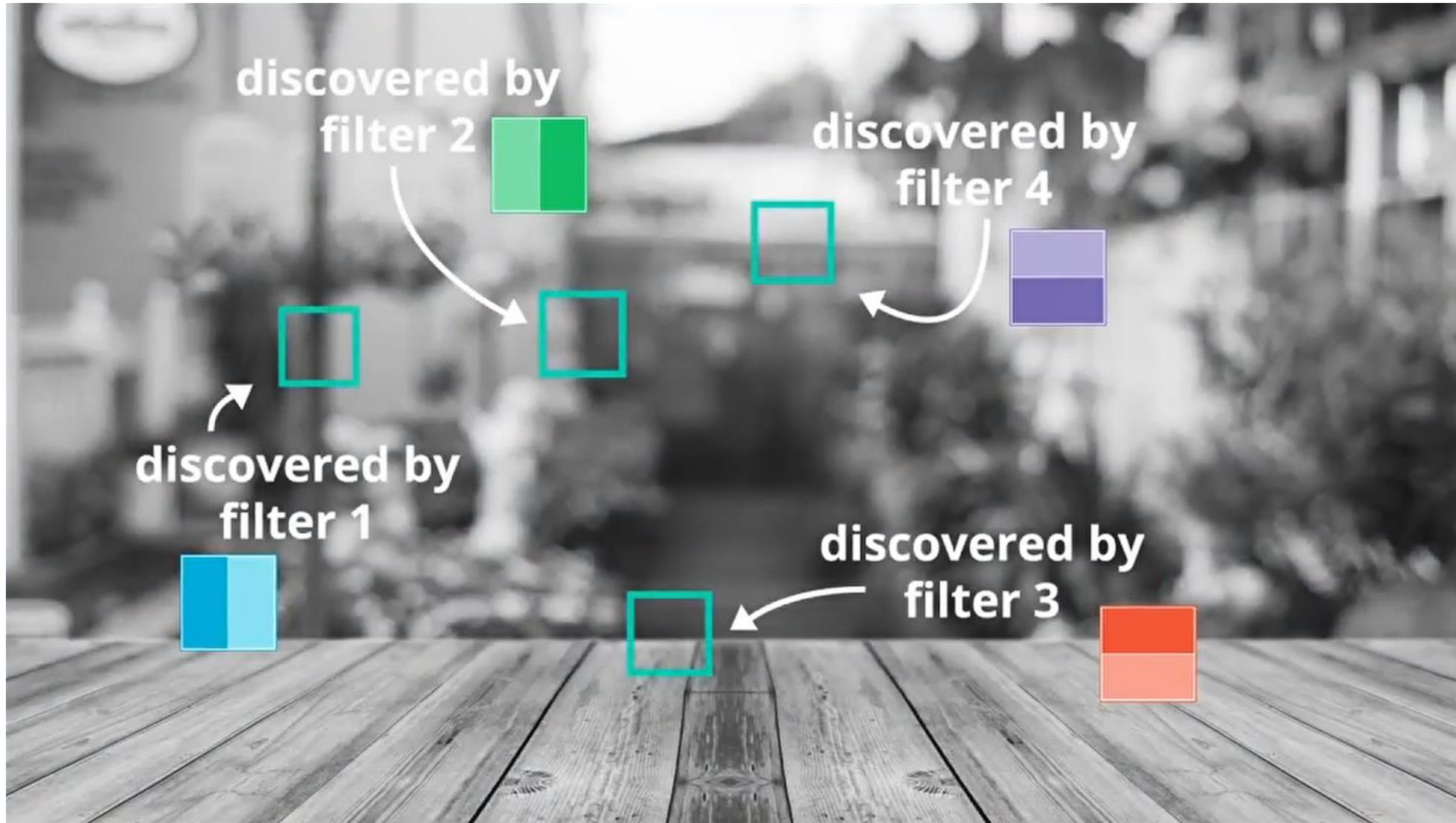
Feature map



Feature map



Different Filter, Different information



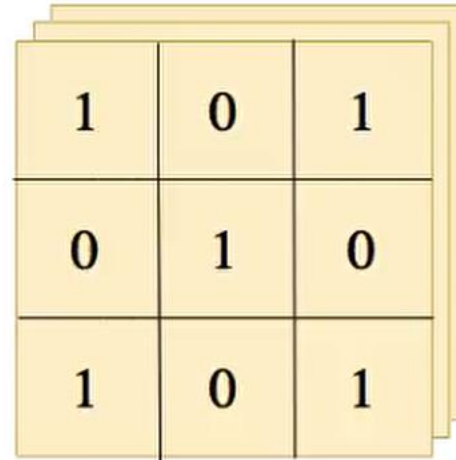
Conv Layer

Convolution Layer



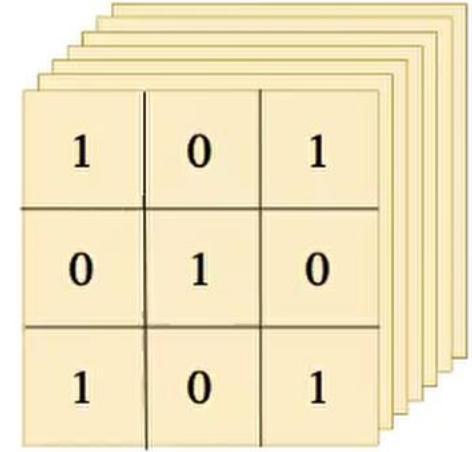
Image
 $1024 \times 682 \times 3$

*



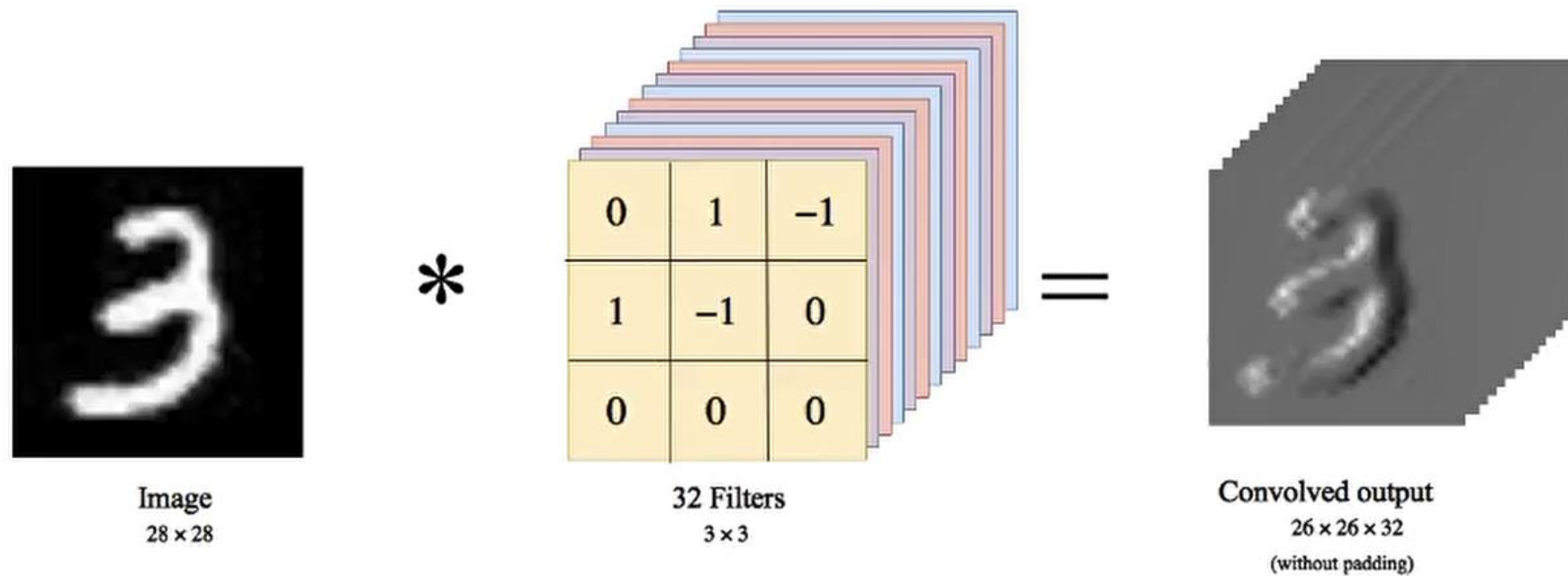
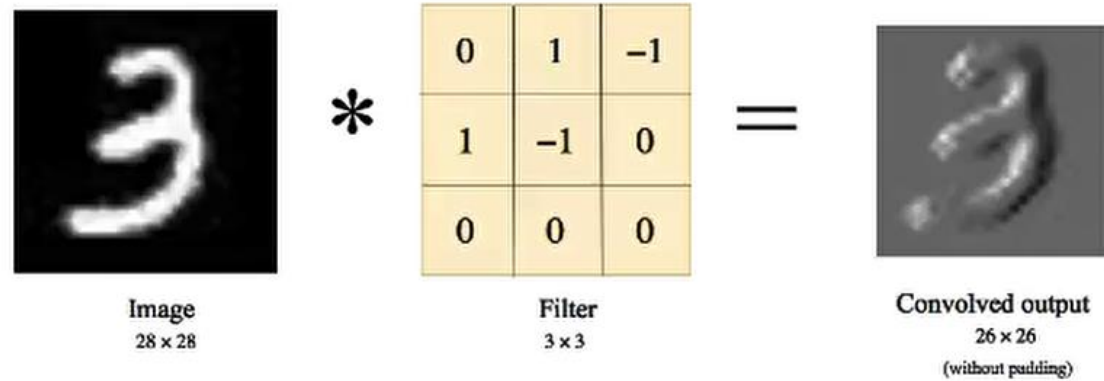
Filter
 $3 \times 3 \times 3$

=



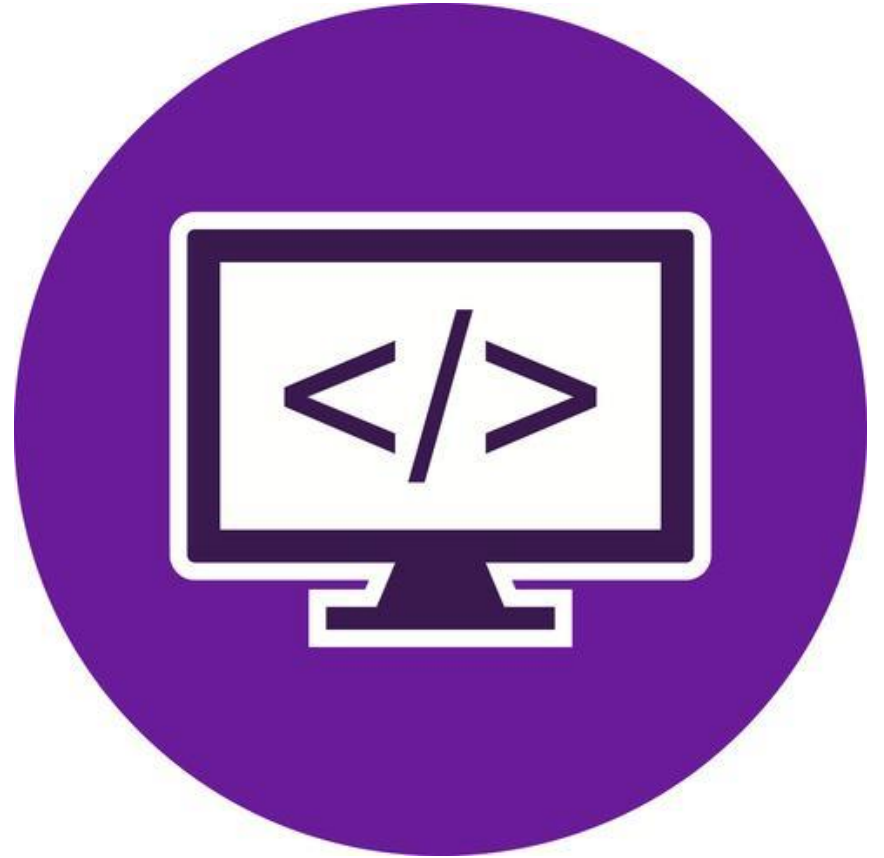
Convolved 2D output
 1022×680
(without padding)

Conv Layer



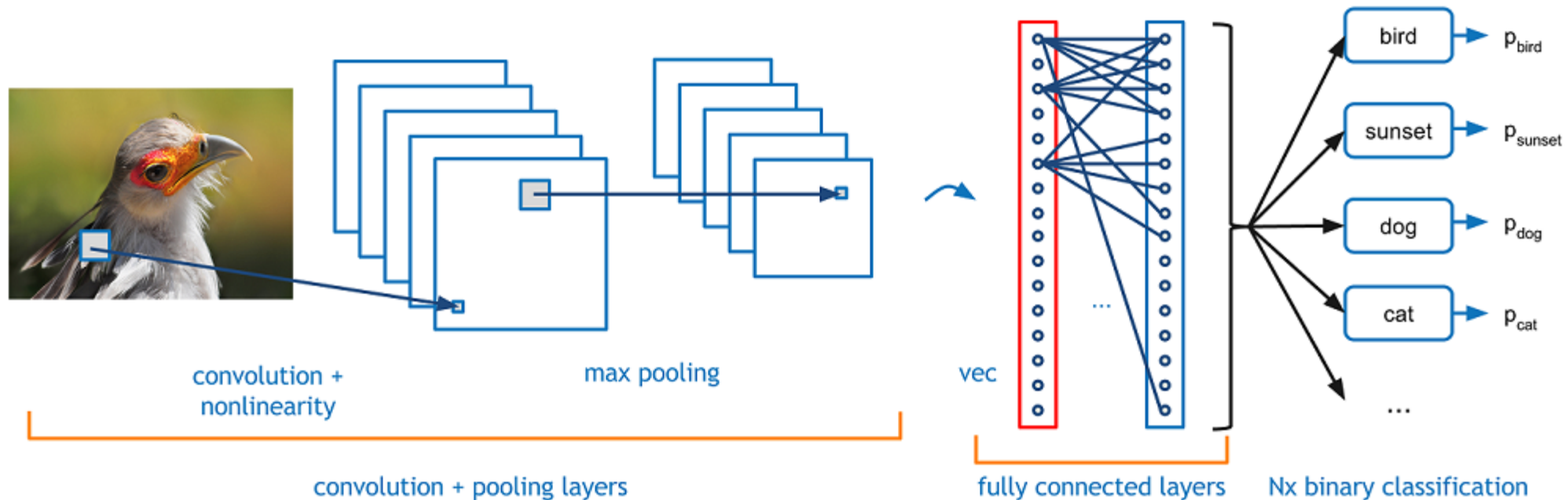
Coding Session

- Visualizing a Convolutional Layer



CNN Layers

- Conv layer
- Fully connected layer



Subsampling

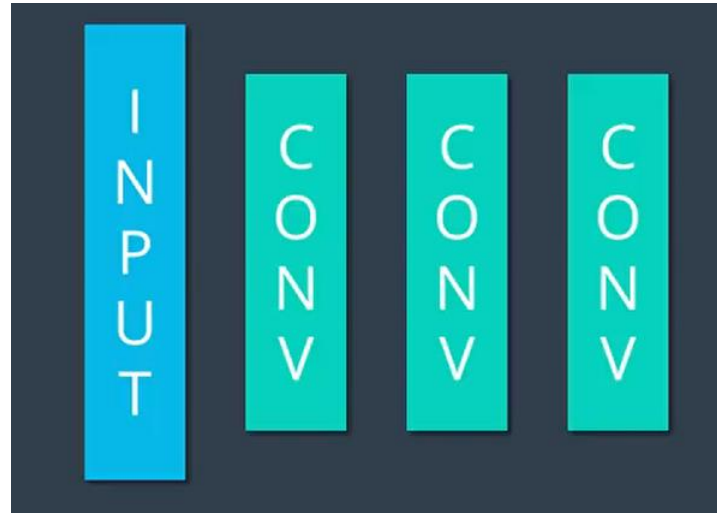


We can subsample the pixels to make image smaller
fewer parameters to characterize the image

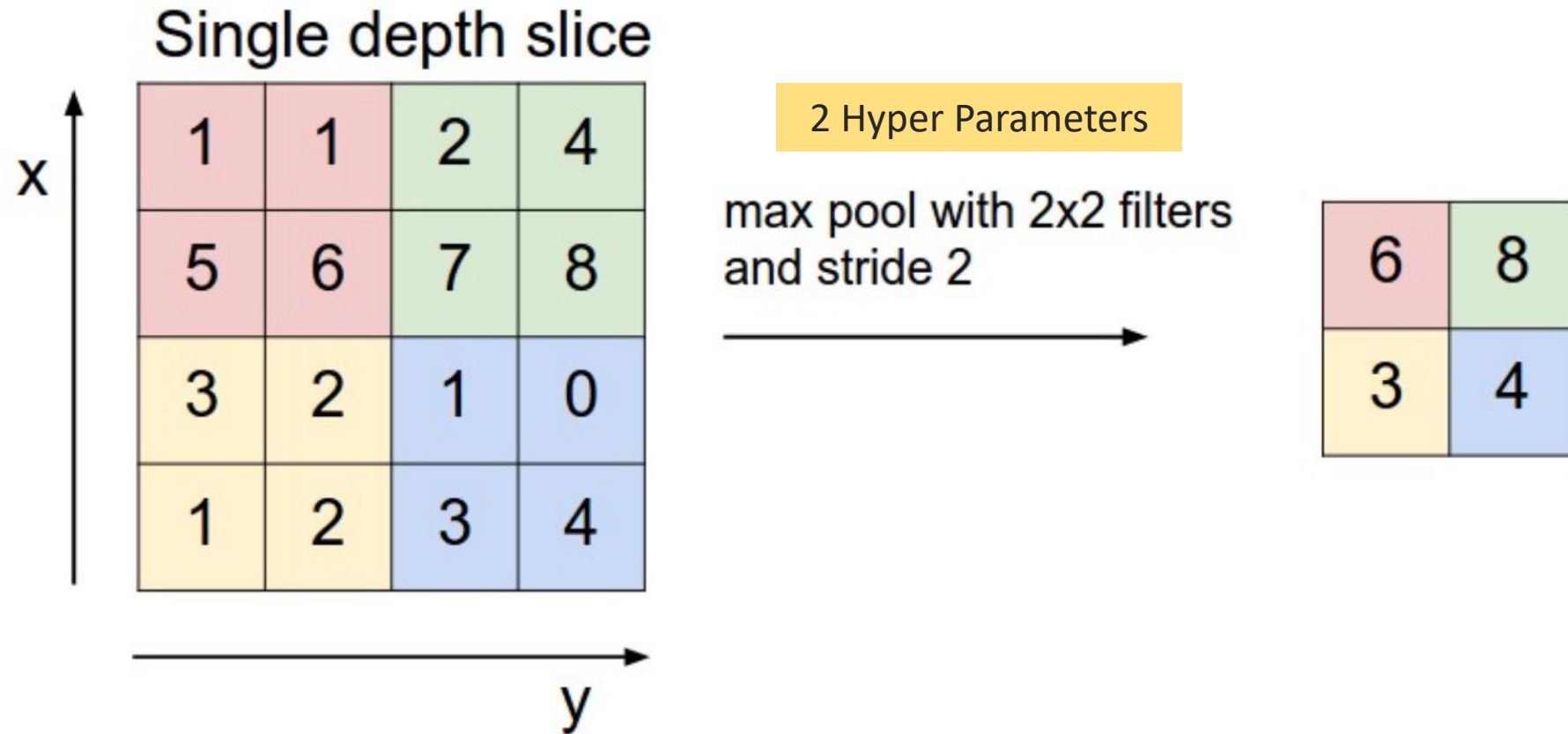
Subsampling



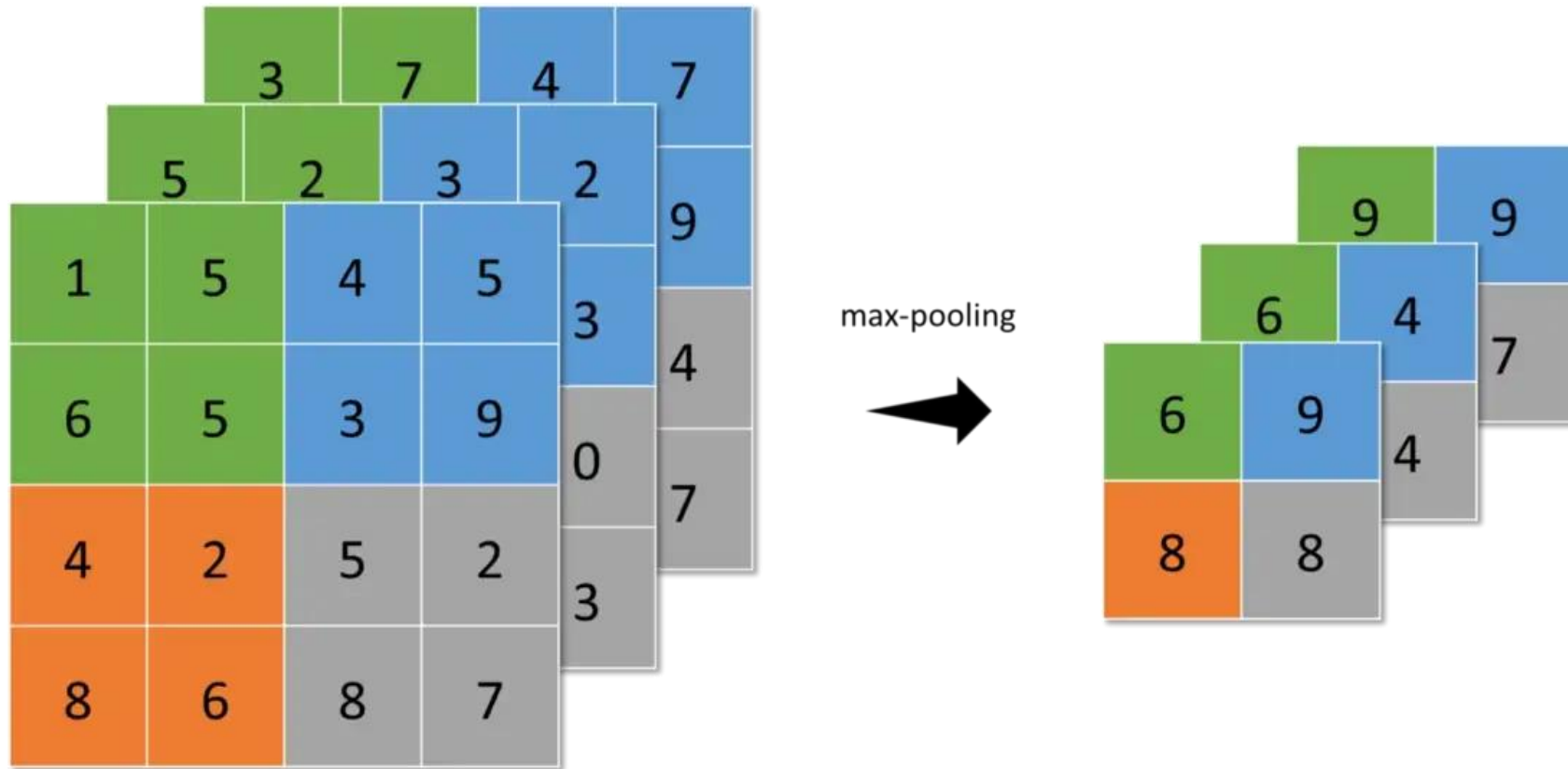
Pooling Layer



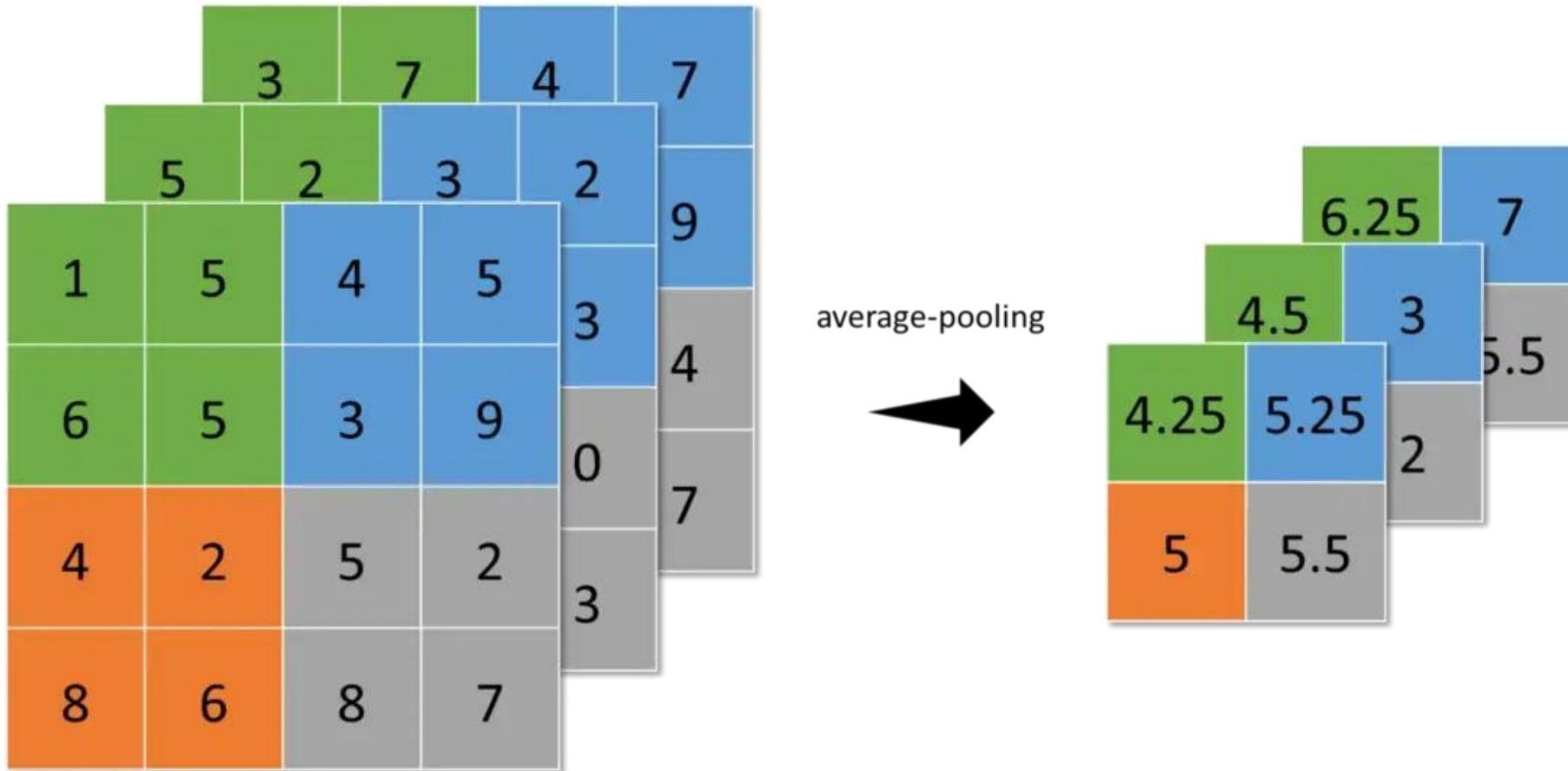
Pooling Layer



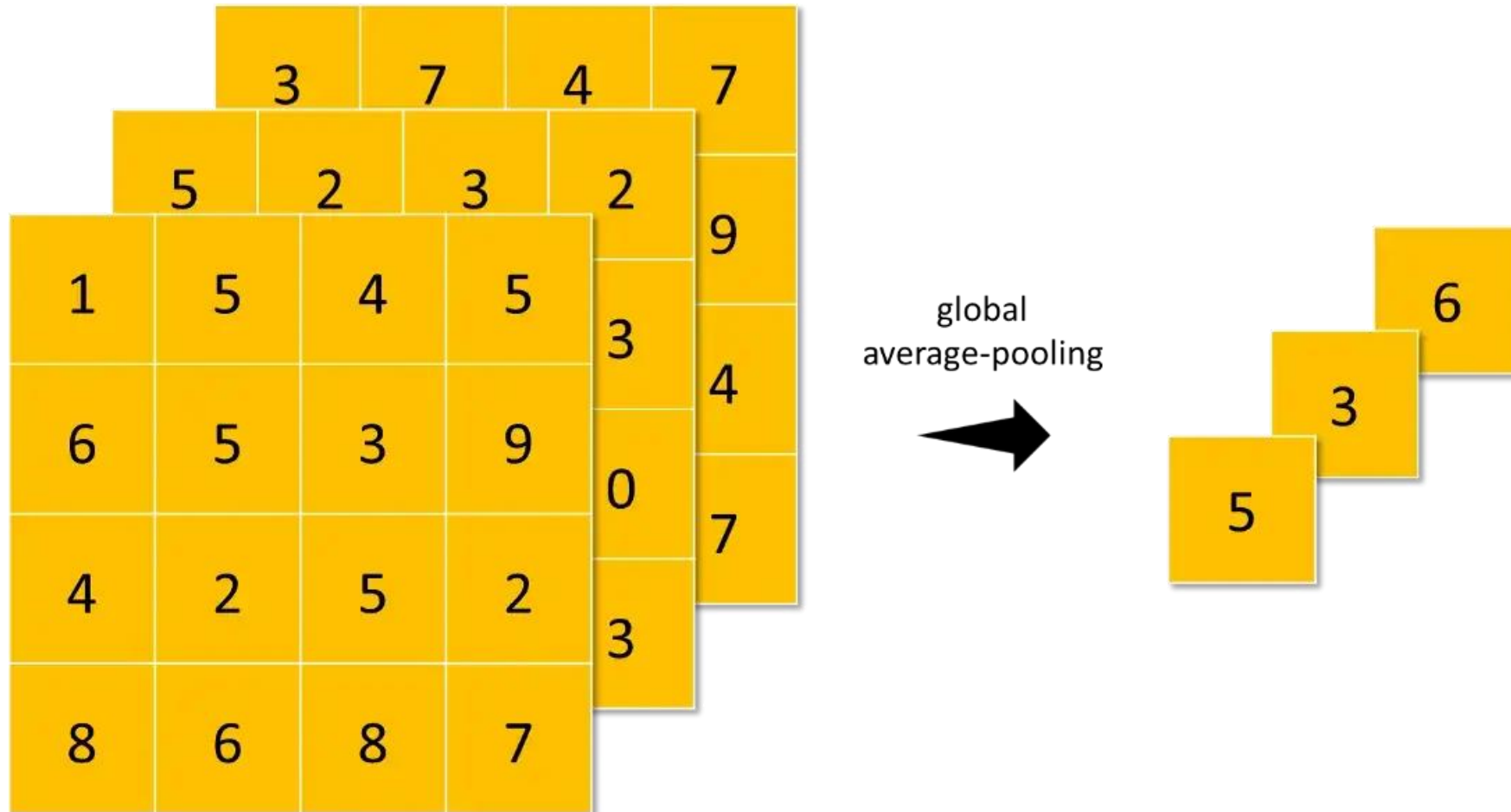
Max Pooling



Average Pooling



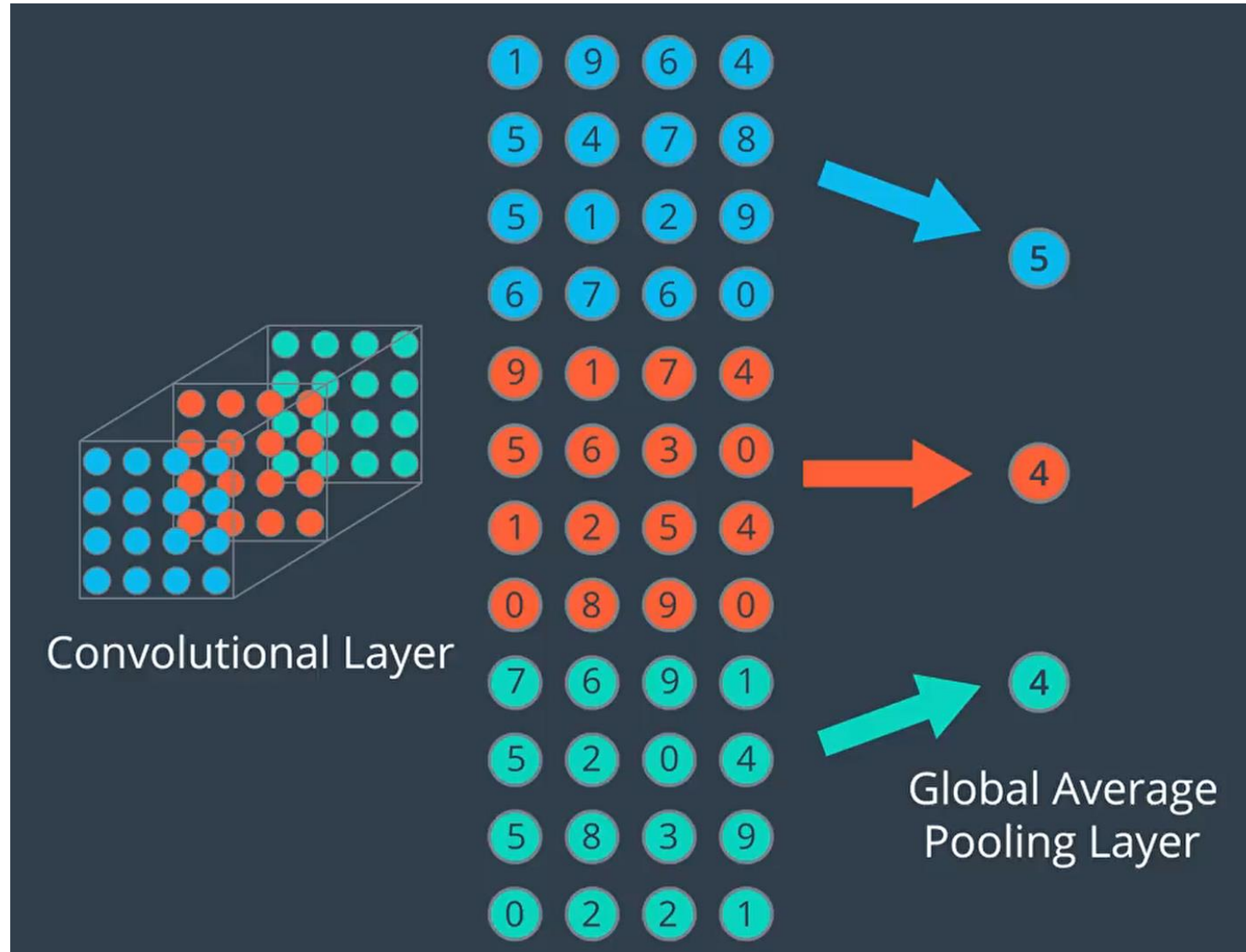
Global Average Pooling



Pooling

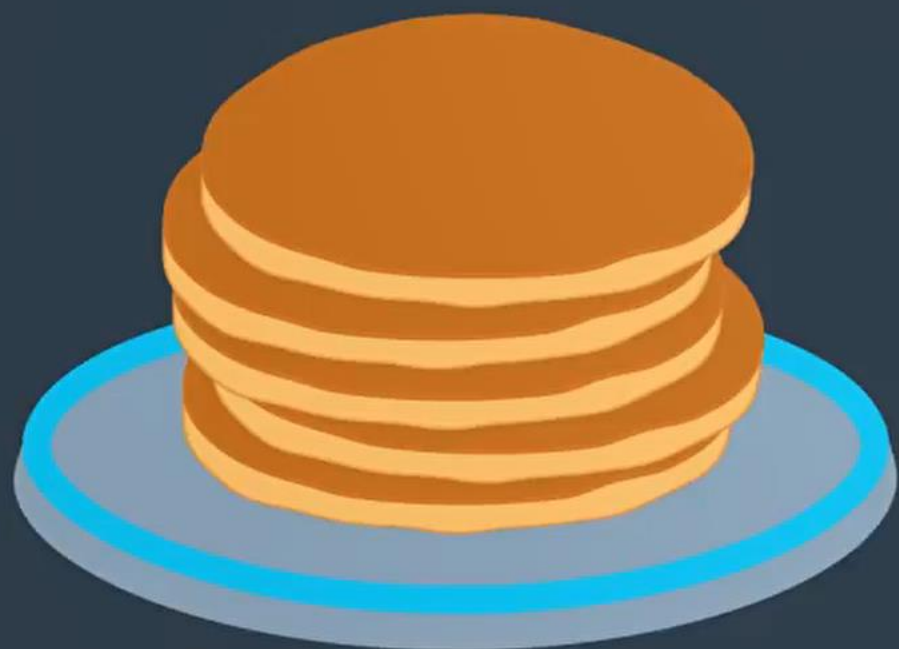


Pooling



Pooling





Convolutional Layer

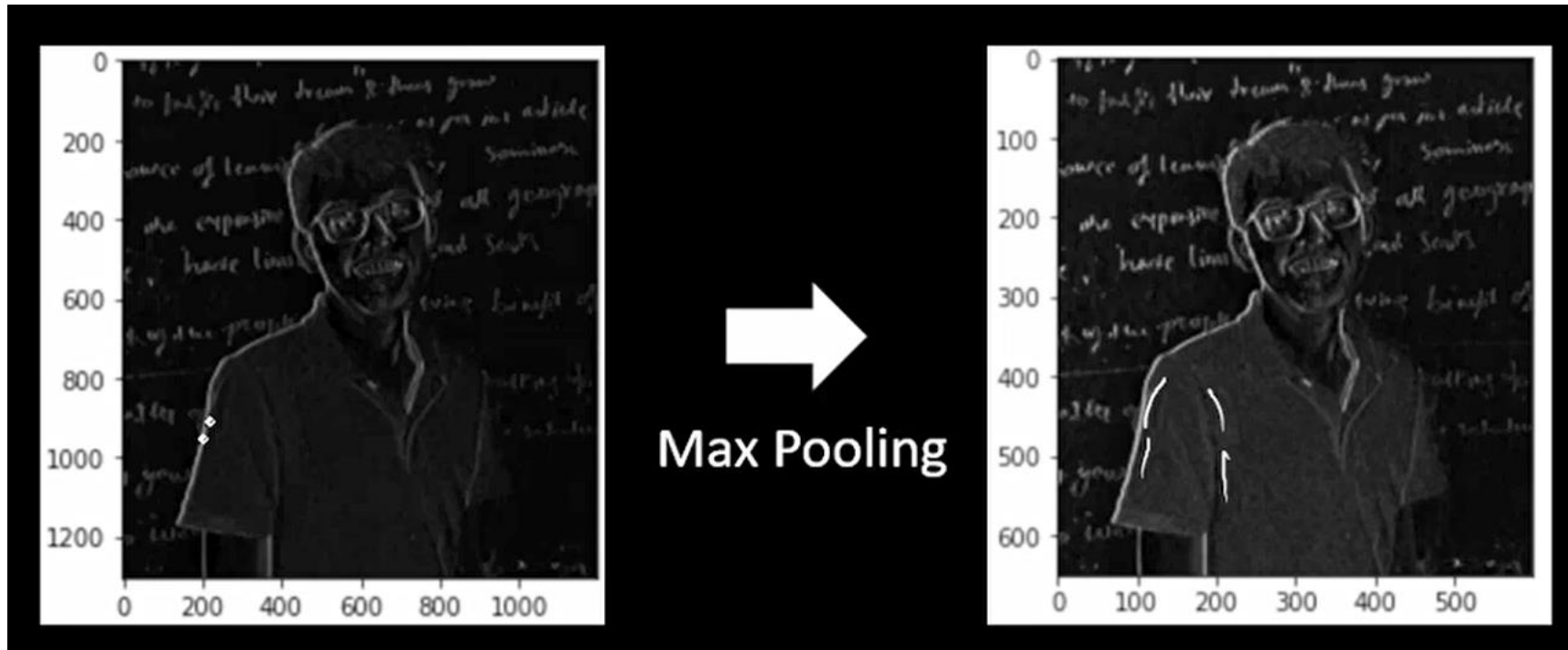


Pooling Layer



Global Pooling Layer

Enhanced Feature By Pooling



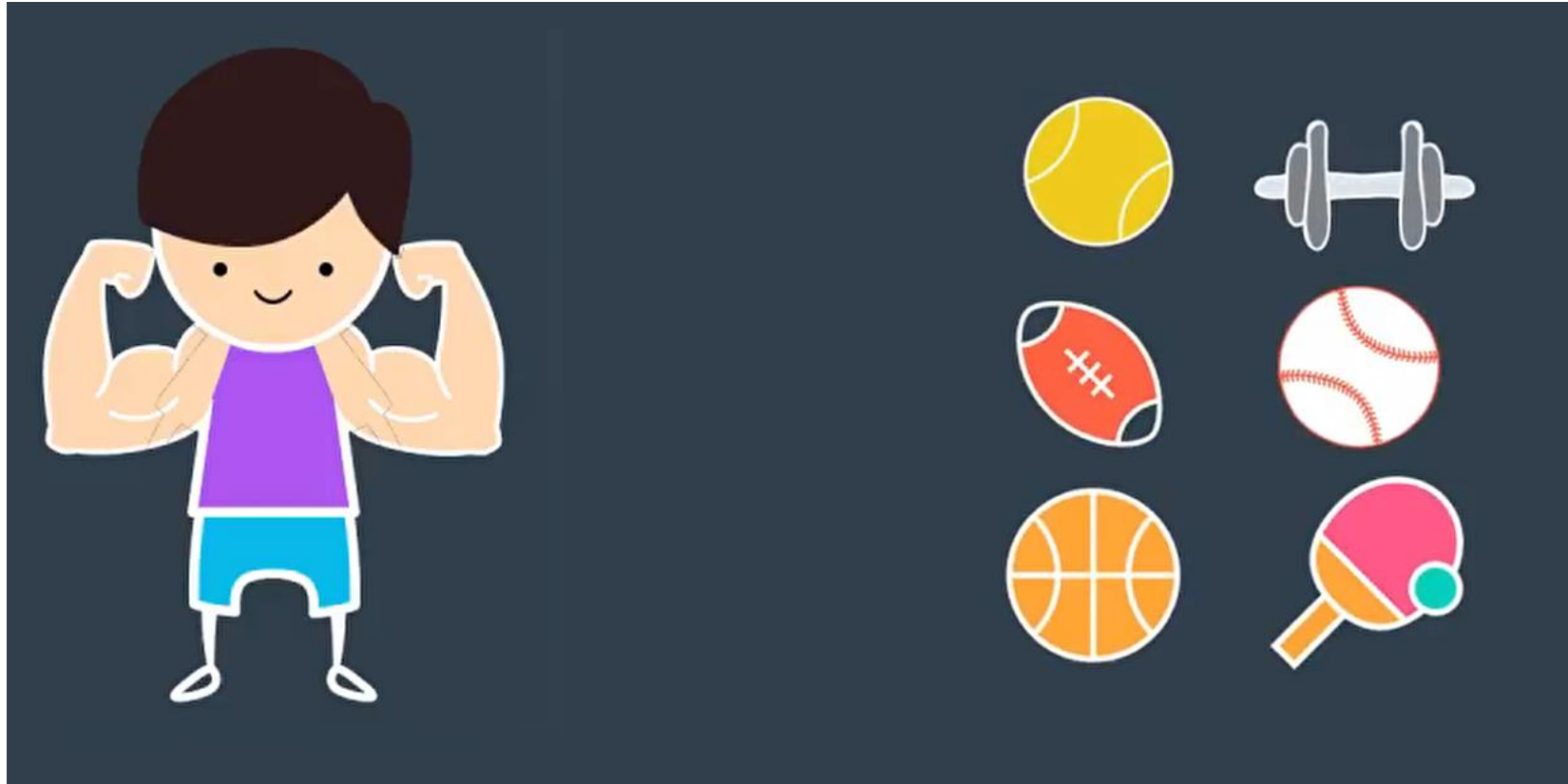
Summary

- Why?
 - Reduce img size -> Reduce Computational cost -> Faster training
 - Enhanced the feature of the img
 - Max pooling introduces invariance
 - Check out figure 8 and 9 on page 337 of the Deep Learning textbook: <http://www.deeplearningbook.org/contents/convnets.html>

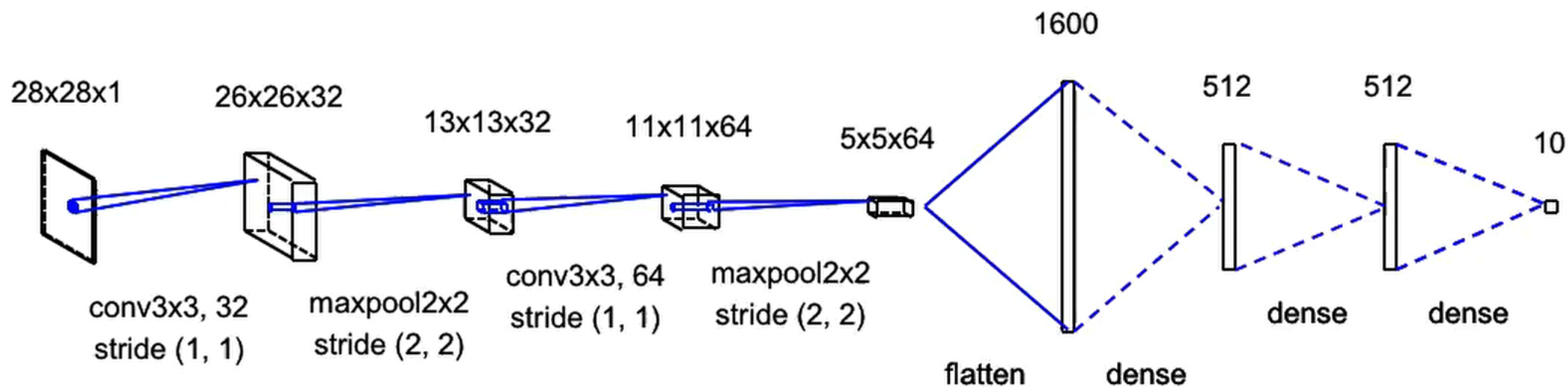
Summary

- Where?
 - After Conv layer(s)
- Other points
 - No parameter, No Training
 - Same number of channels

Dropout



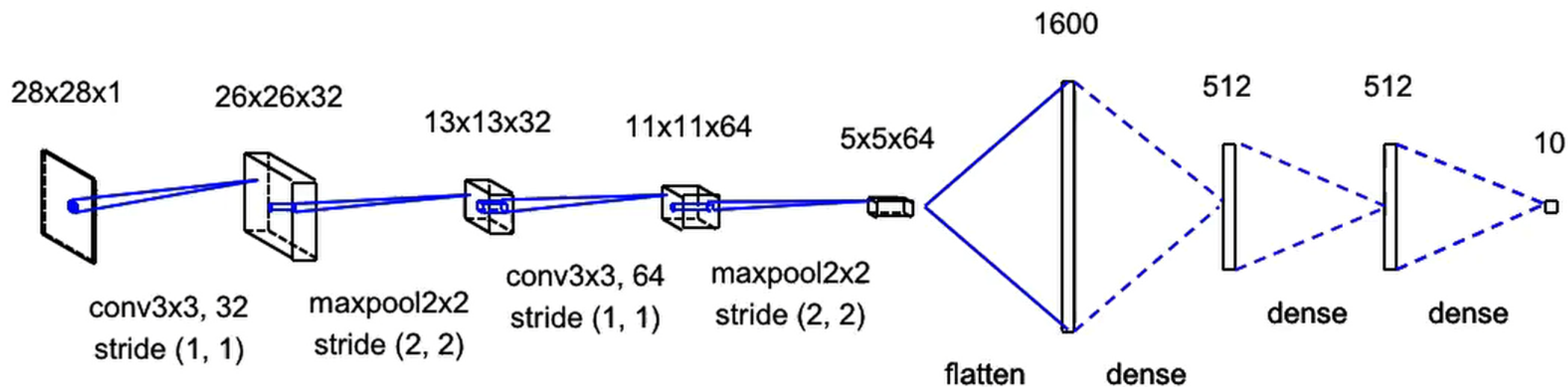
In a nutshell



$$\begin{aligned} Out_{width} &= I_{width} - h_{width} + 1 \\ &= 28 - 3 + 1 = 26 \end{aligned}$$

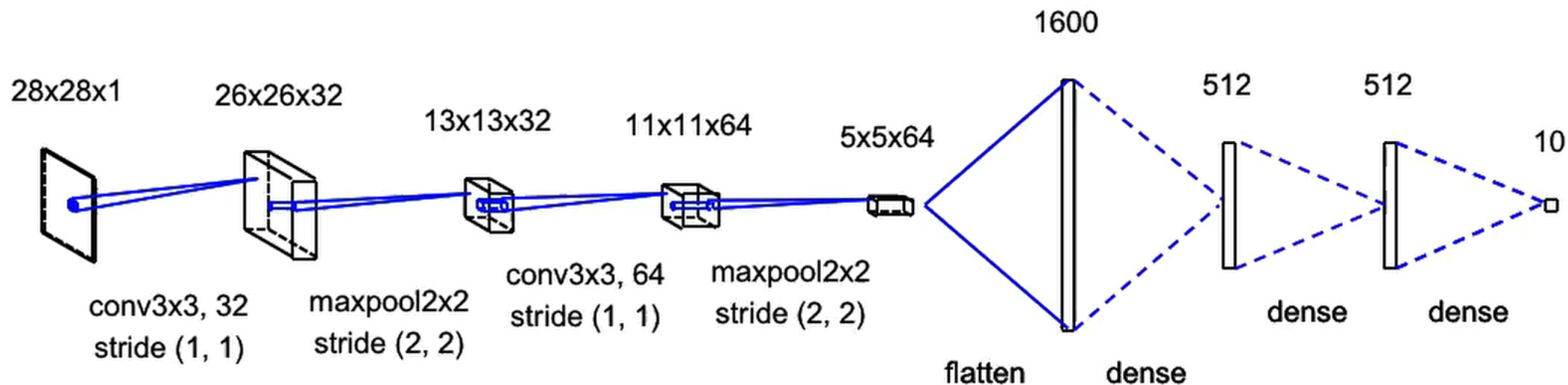
$$Out_{depth} = 32$$

In a nutshell



$$\begin{aligned} Out_{width} &= \frac{I_{width} - h_{width} + 1}{stride} \\ &= \frac{26 - 2 + 1}{2} = [13.5] = 13 \end{aligned}$$

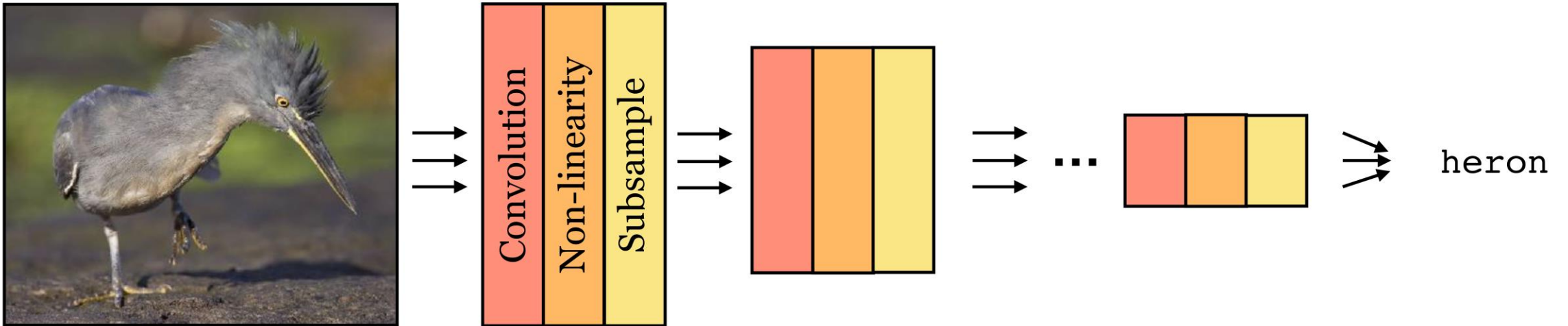
In a nutshell



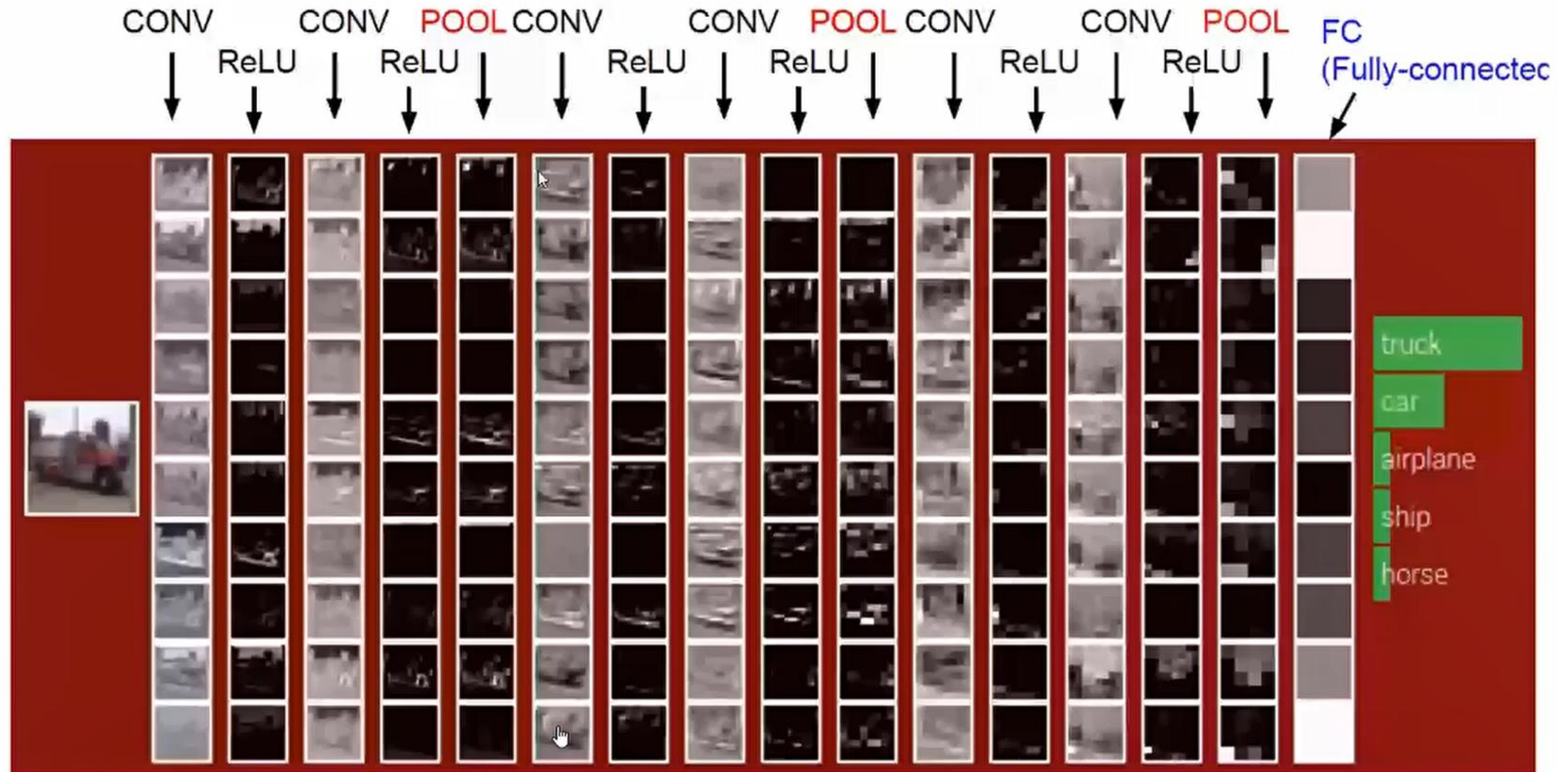
Keras: Padding parameter

1. Valid: No padding for Image border
2. Same: Padding for Image border

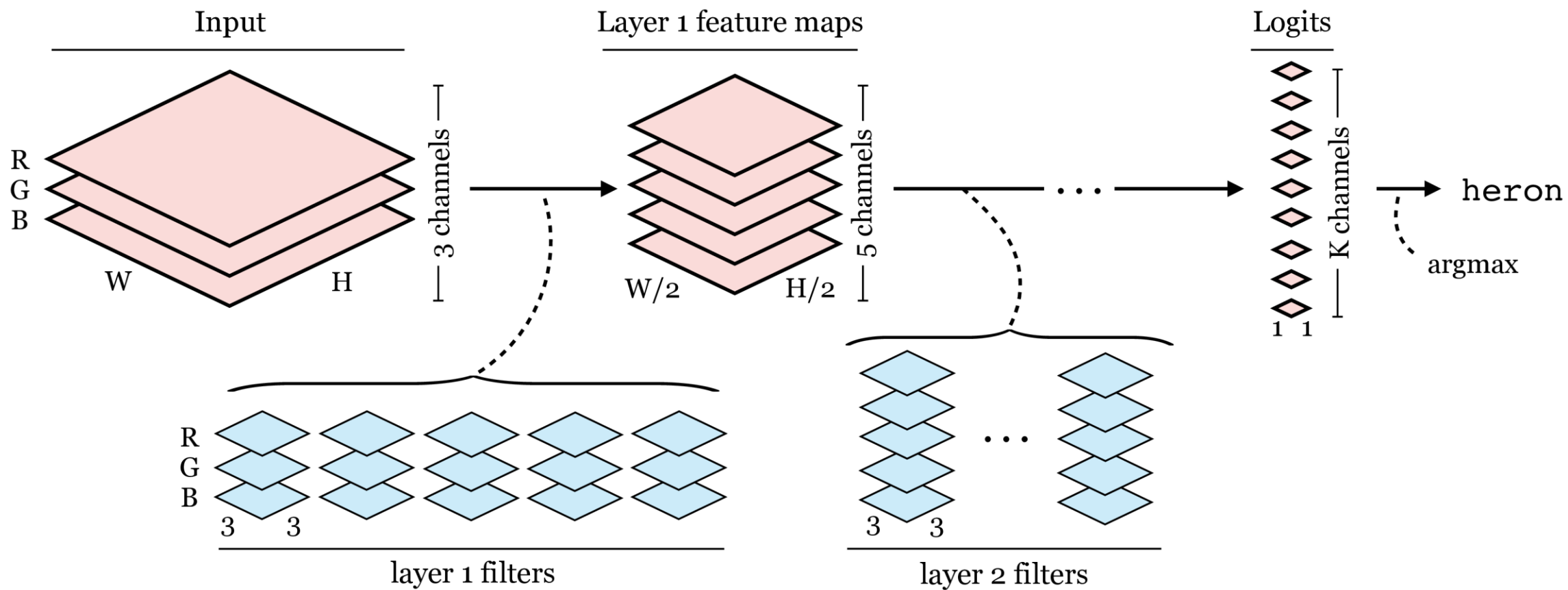
Recap



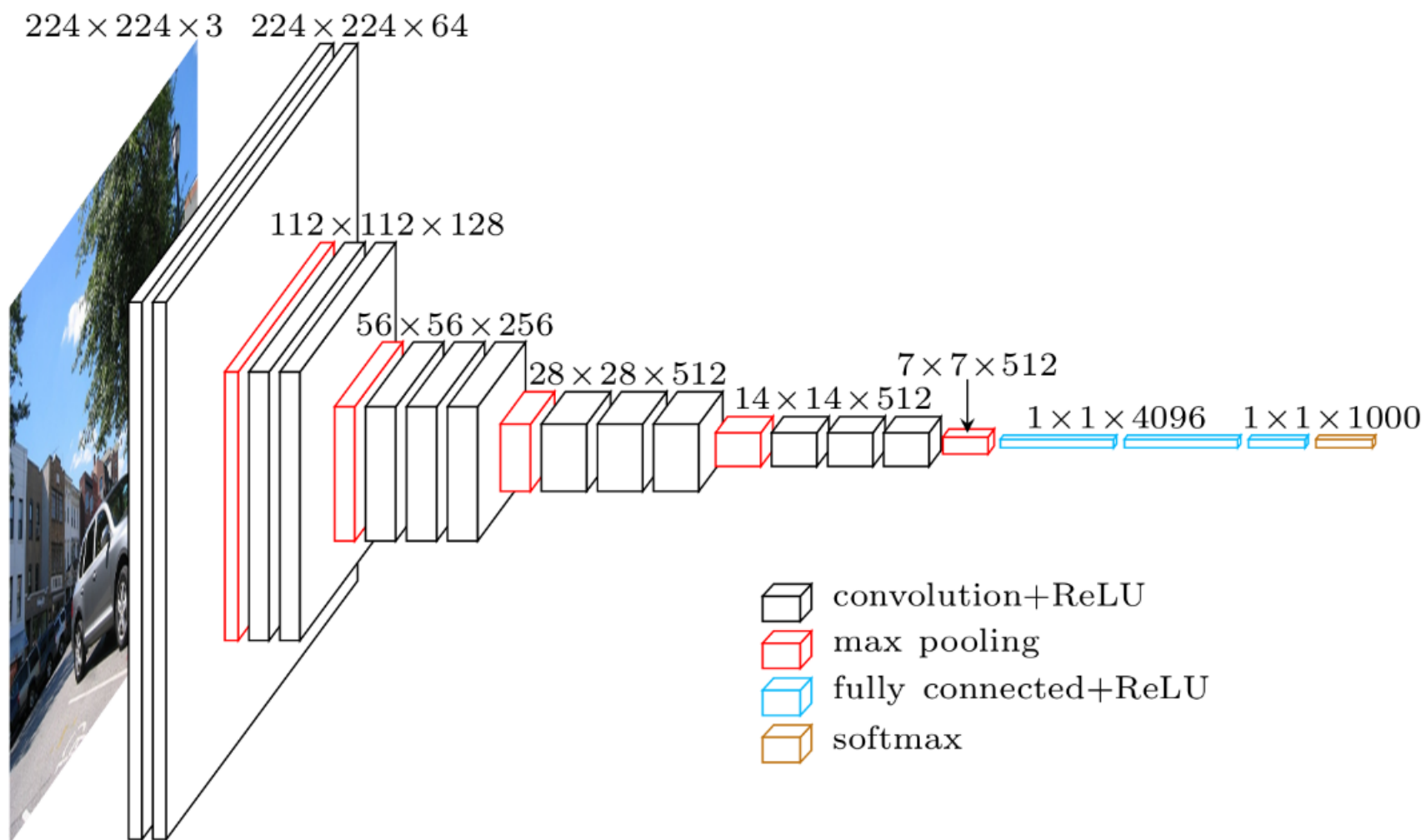
Recap



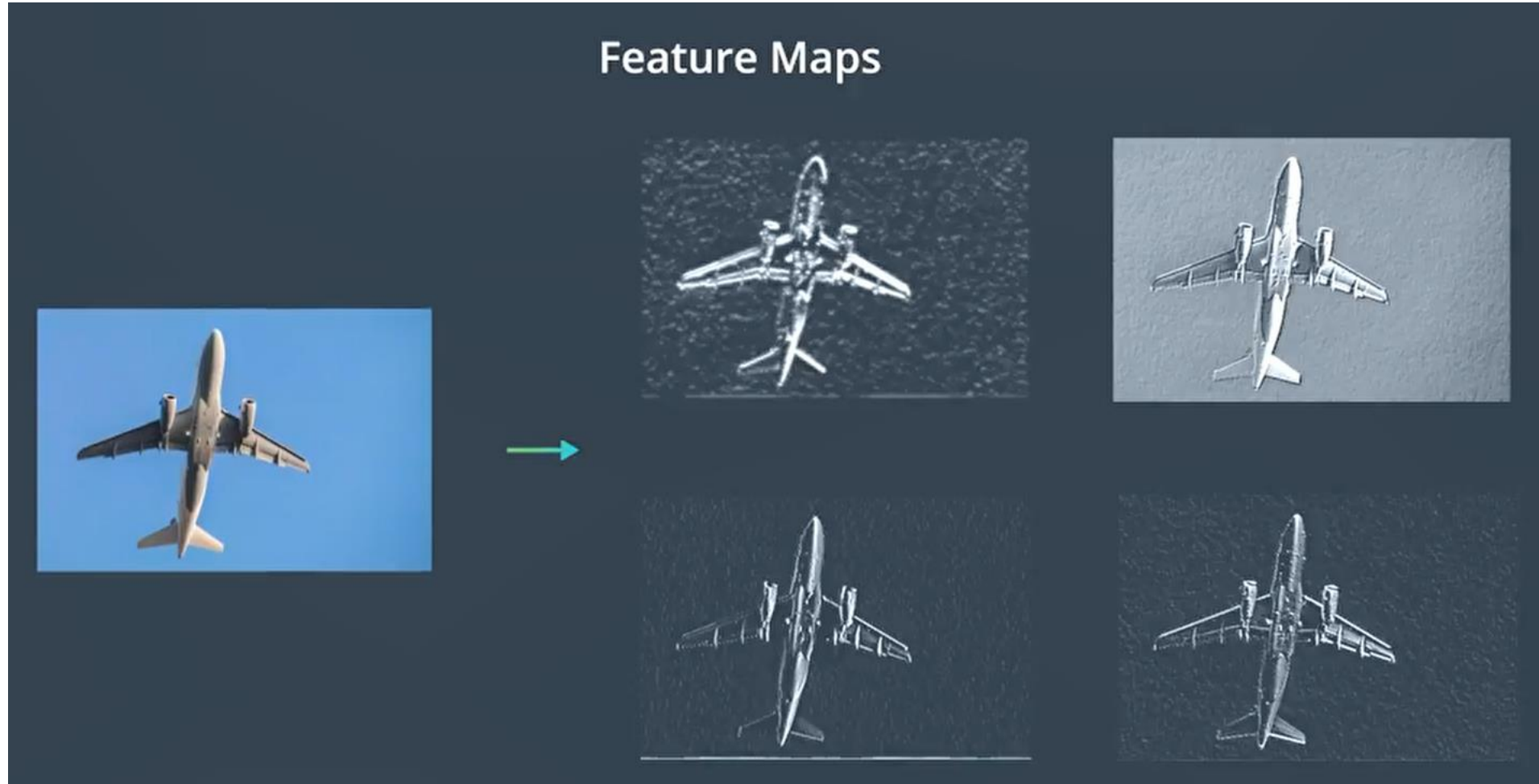
Recap



In a nutshell

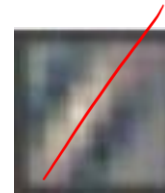
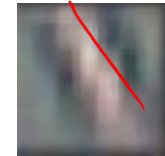
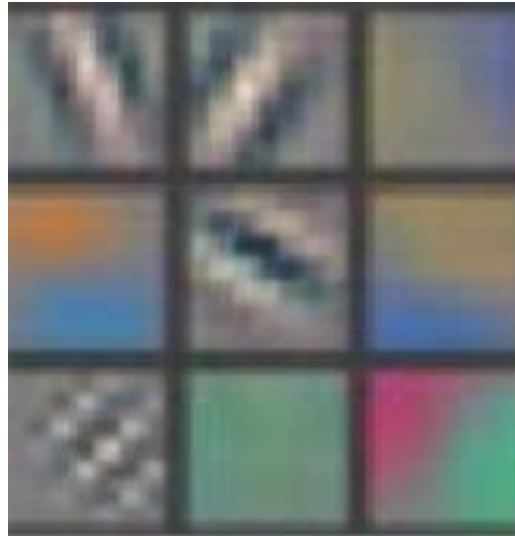


Feature map visualization

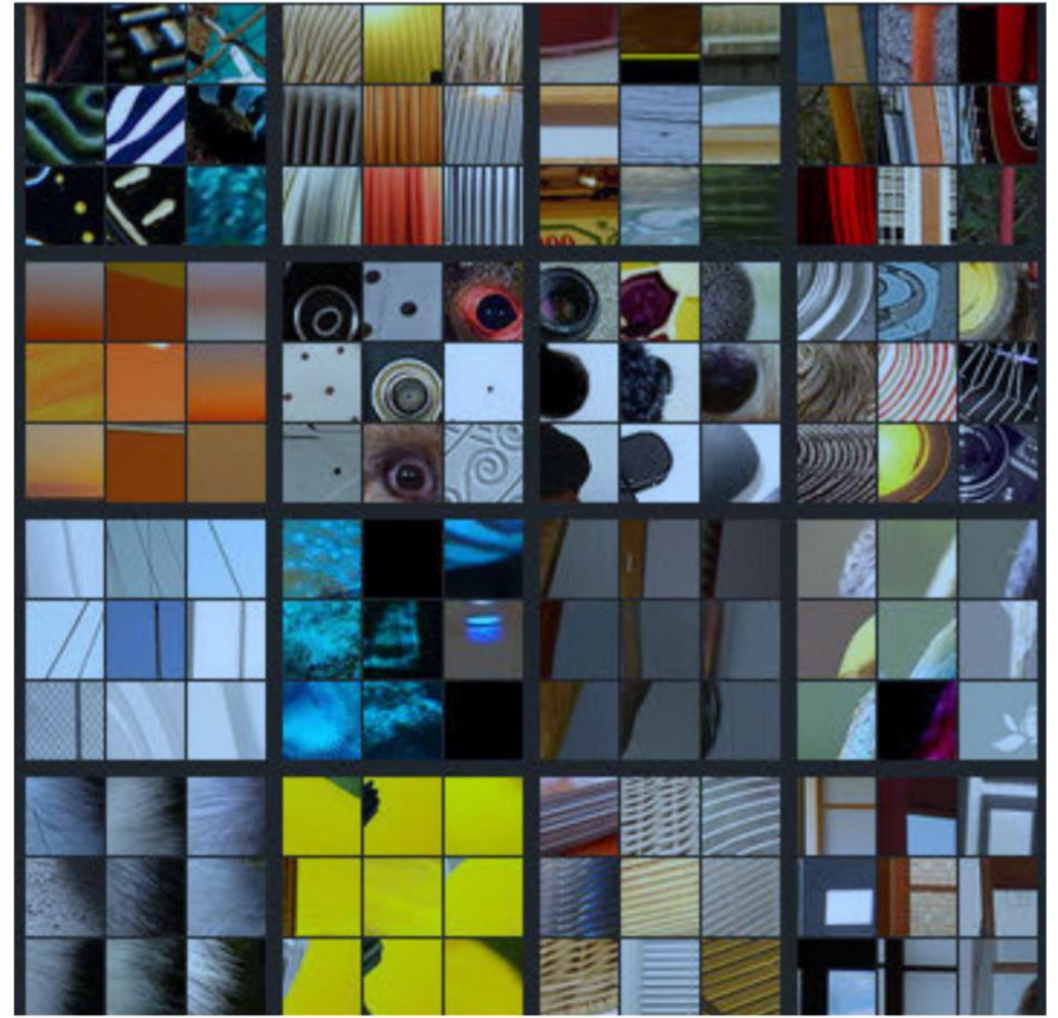
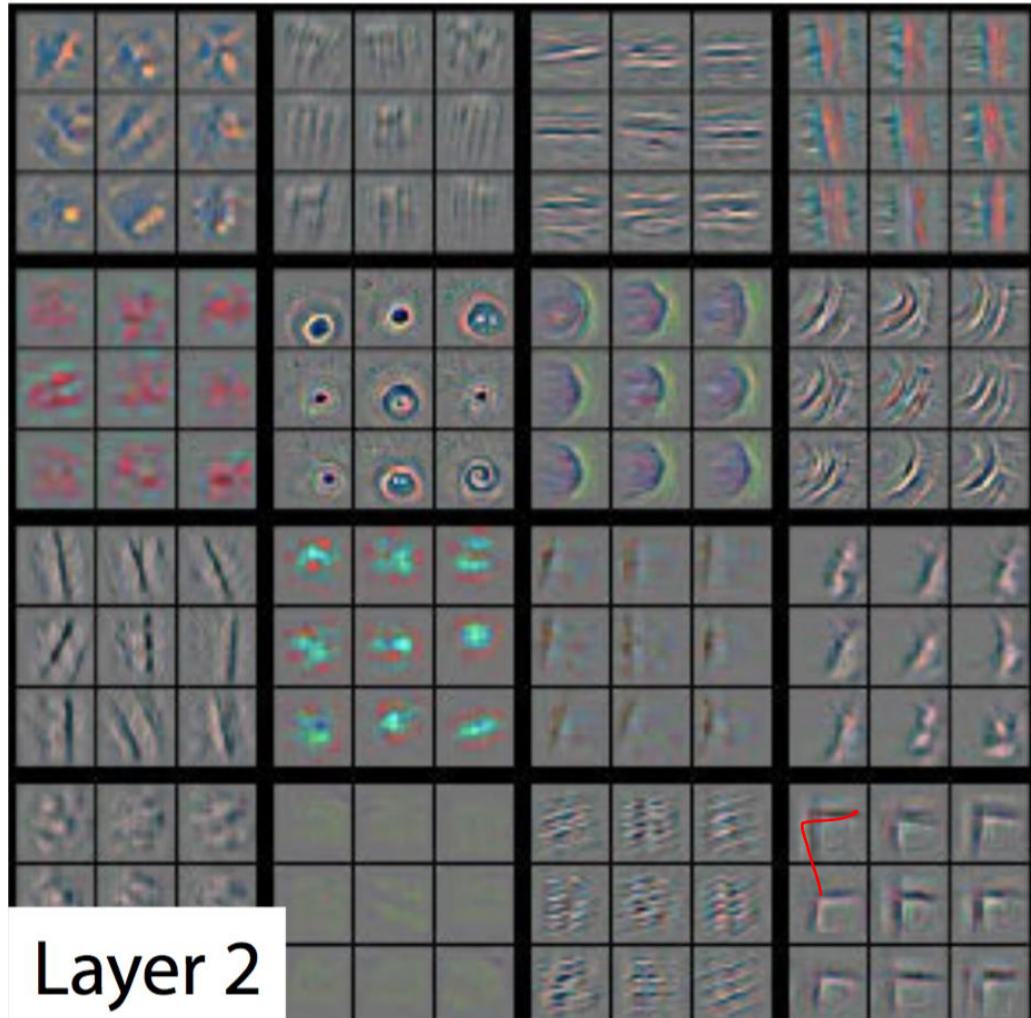


Feature map visualization

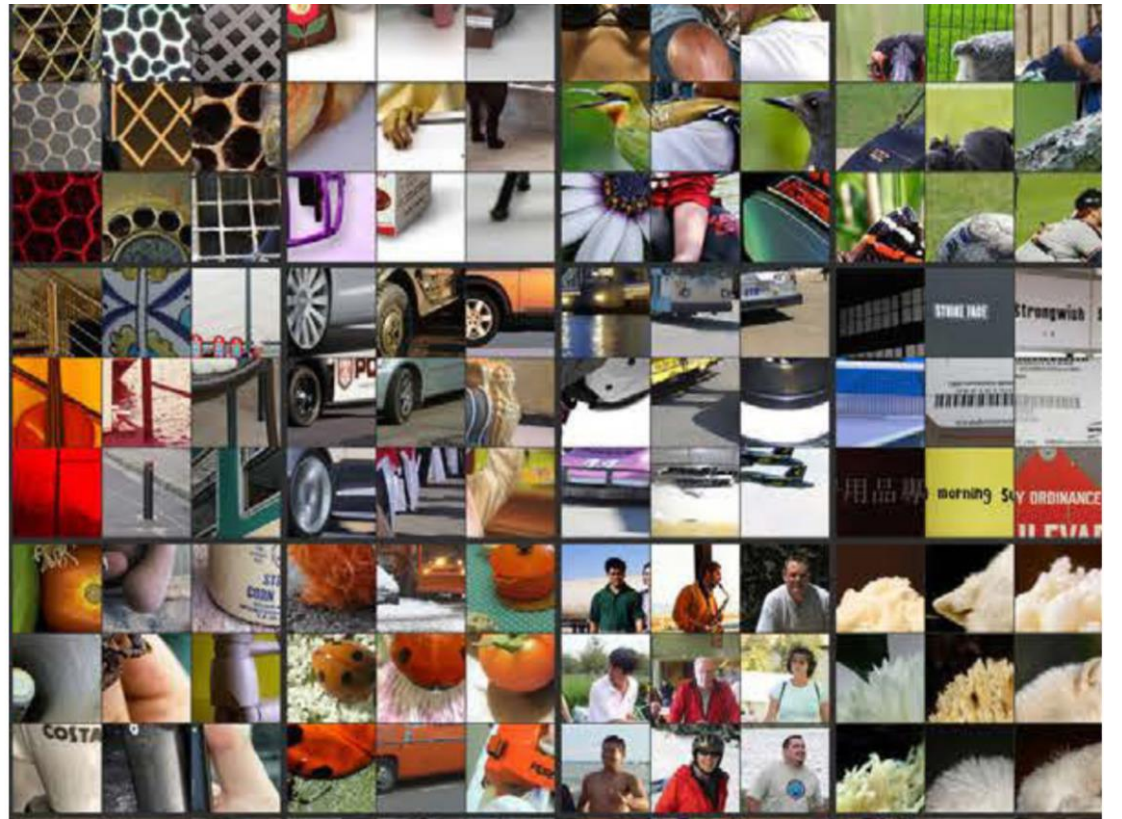
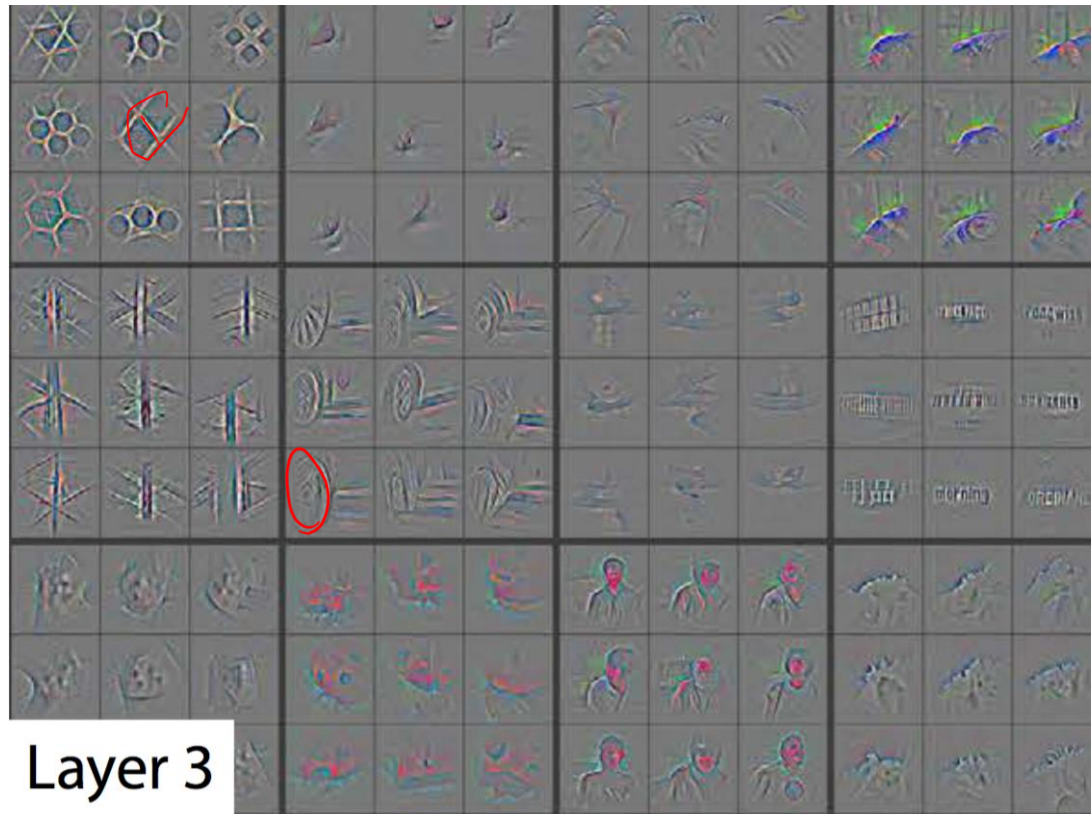
Layer 1



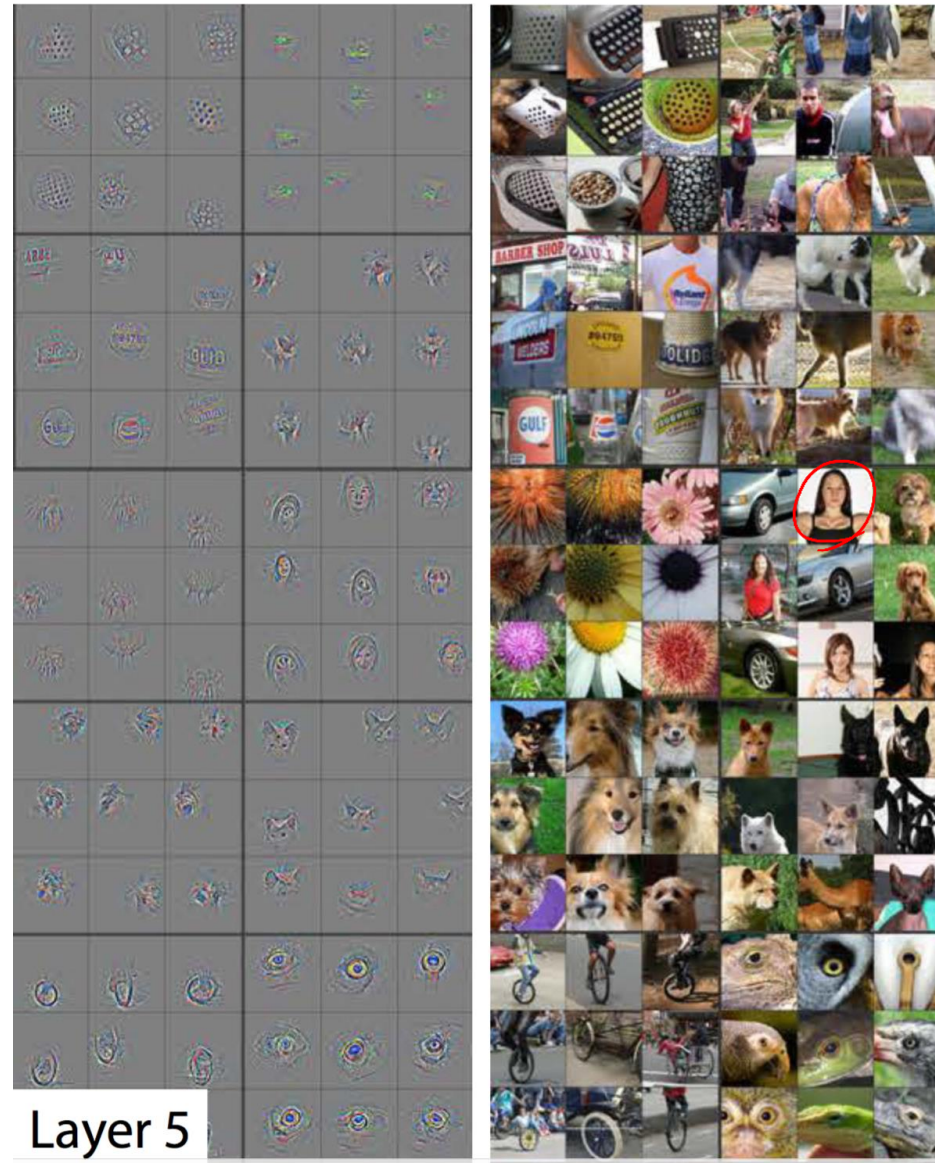
Feature map visualization



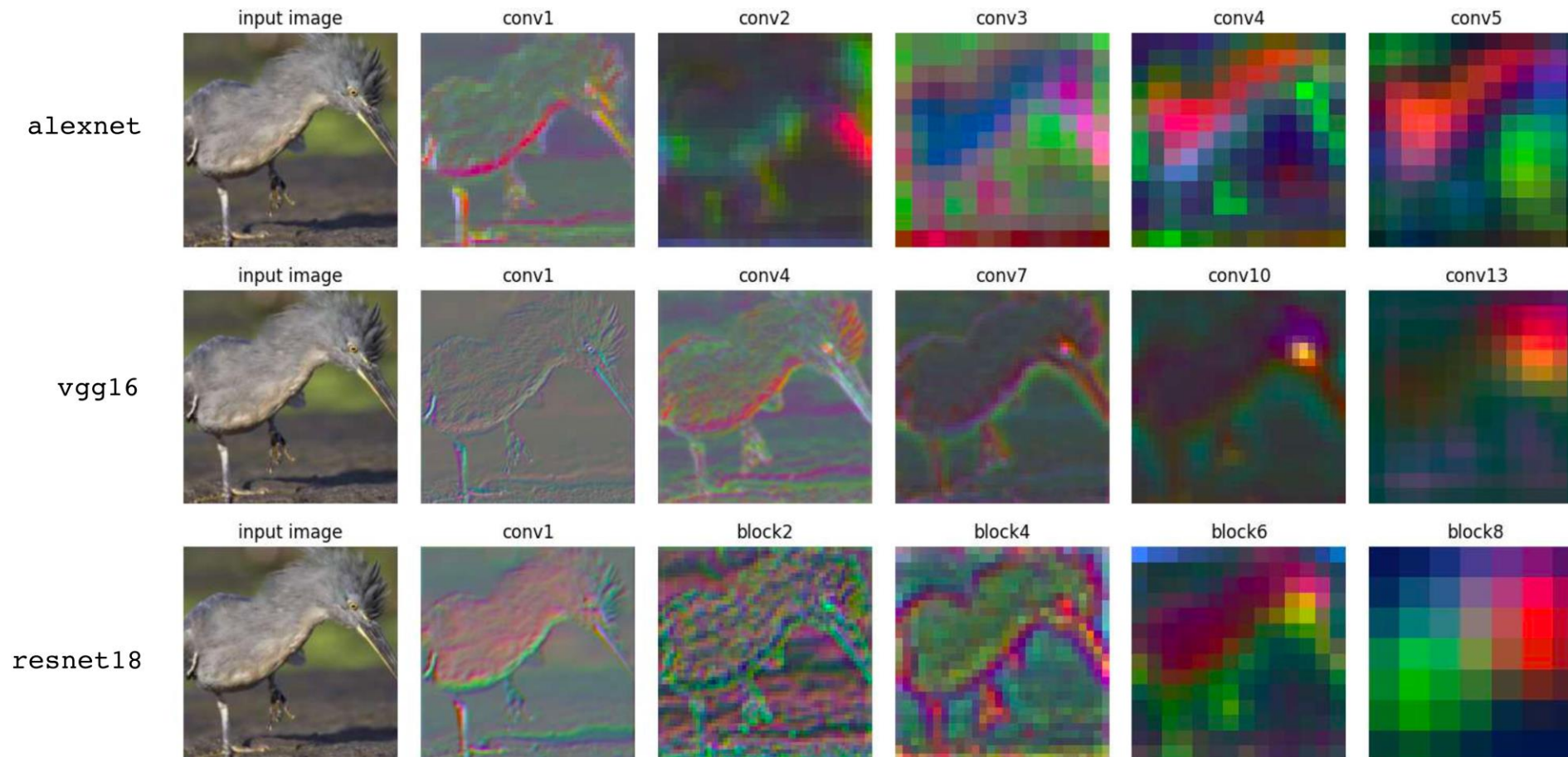
Feature map visualization



Feature map visualization

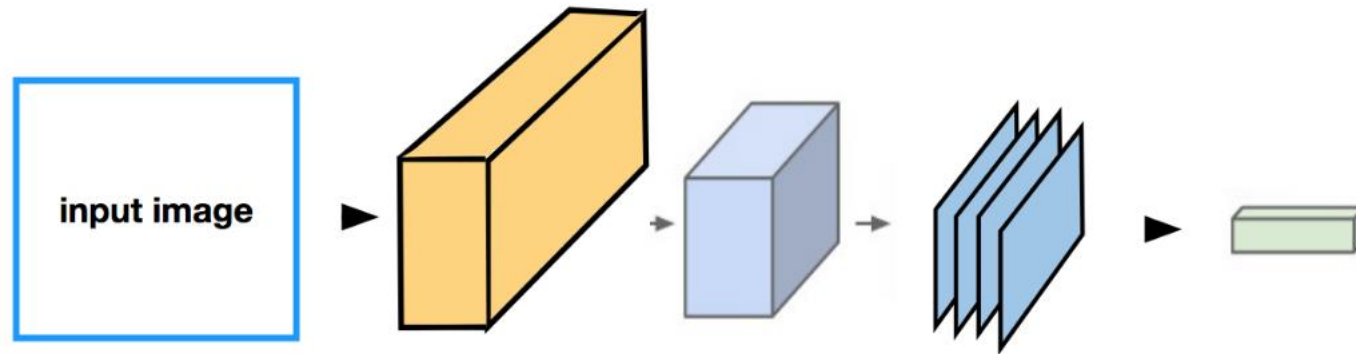


Feature map visualization

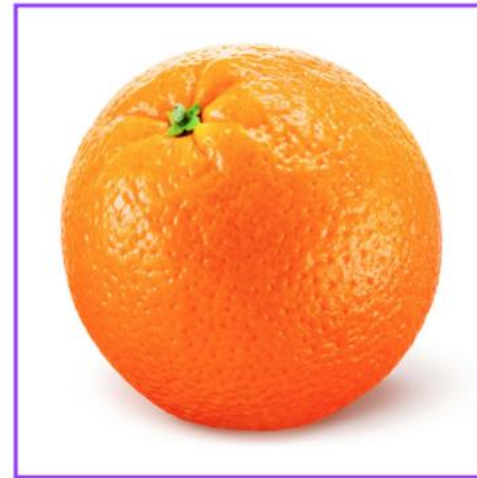


Last Feature Vector and t-SNE

how can we understand what's going on in this final feature vector?
What kind of information has it distilled from an image?



Nearest neighbors in pixel space



Nearest neighbors in feature space

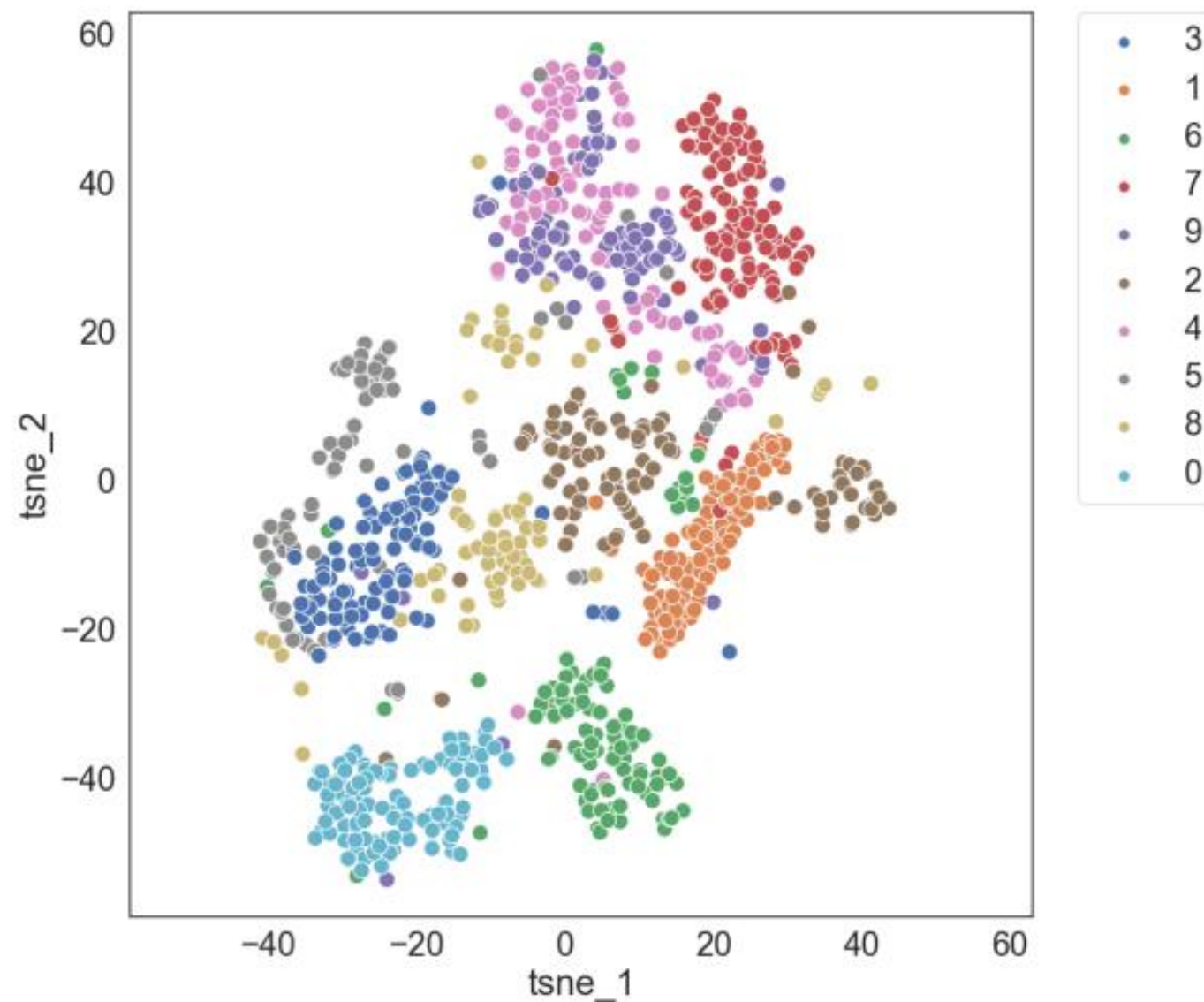


T-SNE

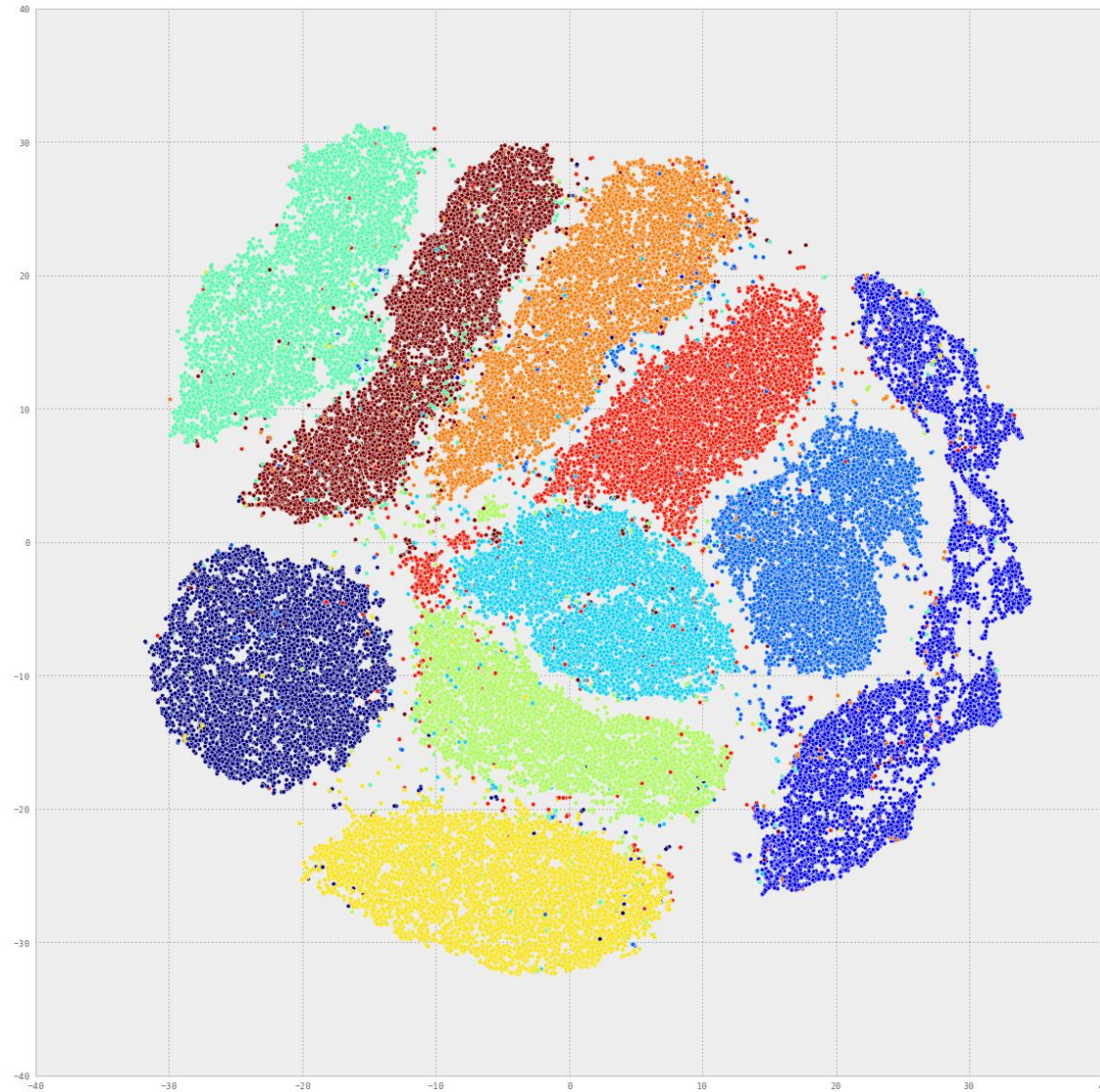
t-distributed stochastic neighbor embeddings

It's a non-linear dimensionality reduction that, again, aims to separate data in a way that clusters similar data close together and separates differing data

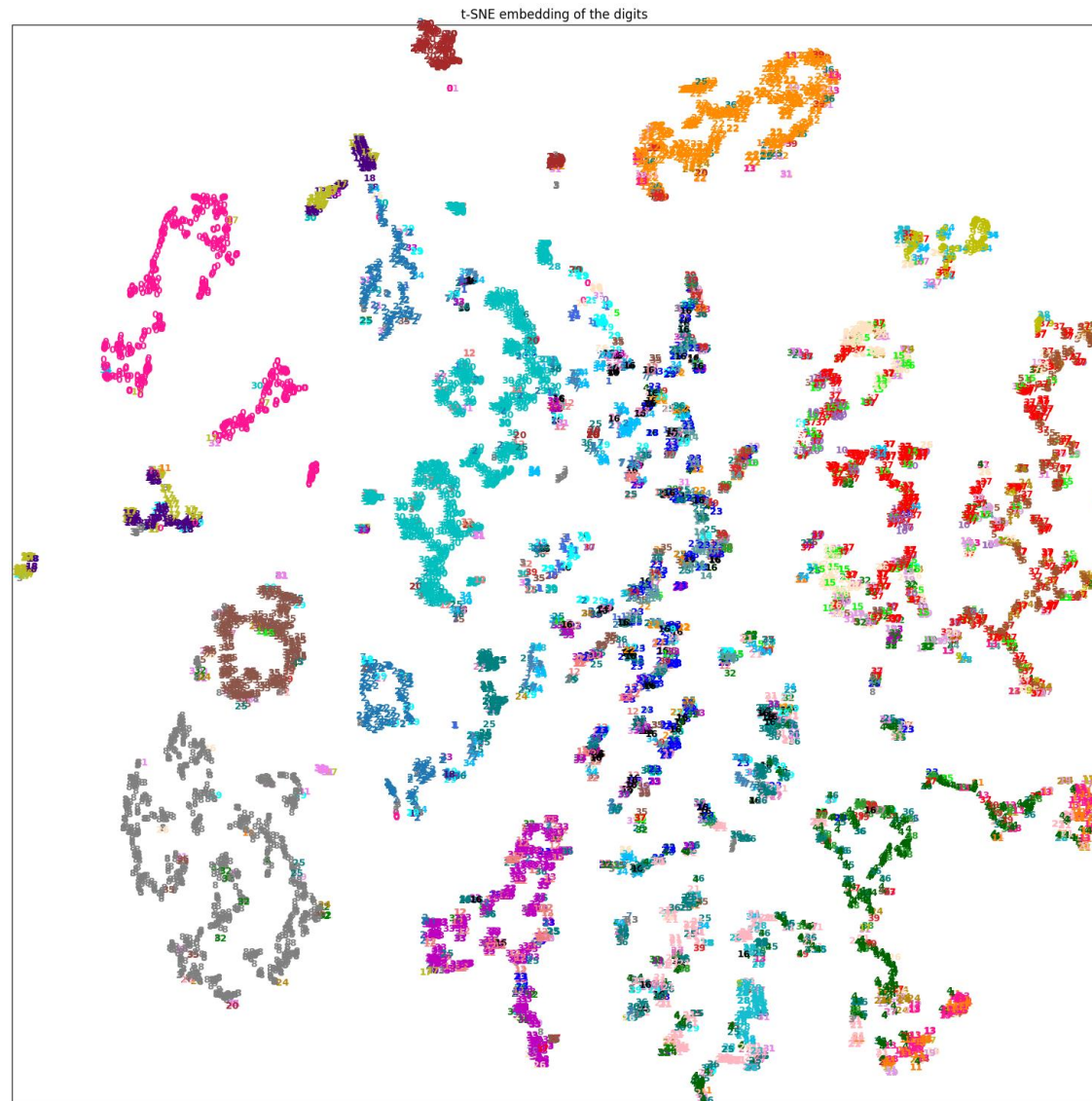
T-SNE (example)



T-SNE (example)



T-SNE (example)



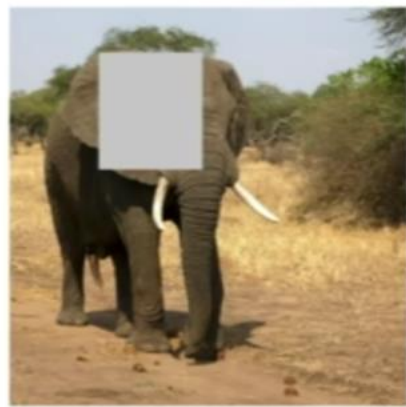
Other Feature Visualization Techniques

- Occlusion Experiments
- Saliency Maps
- Guided Backpropagation

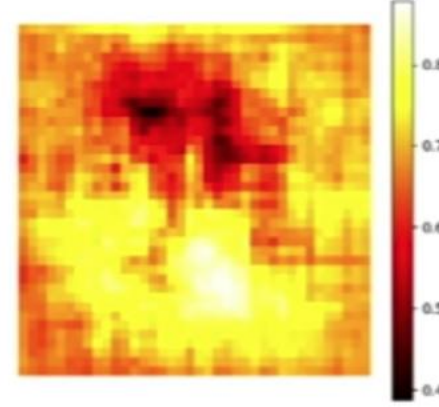
Occlusion Experiments

- The process for an occlusion experiment is as follows:
 - 1.Mask part of an image before feeding it into a trained CNN,
 - 2.Draw a heatmap of class scores for each masked image,
 - 3.Slide the masked area to a different spot and repeat steps 1 and 2.
- The result should be a heatmap that shows the predicted class of an image as a function of which part of an image was occluded. The reasoning is that **if the class score for a partially occluded image is different than the true class, then the occluded area was likely very important!**

Occlusion Experiments



mask part
of an image

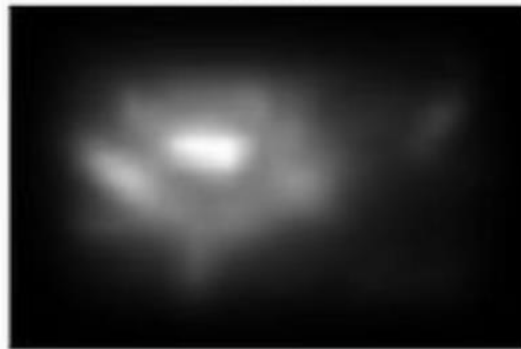


heatmap of class probabilities
as a function of the
masked portion of the image

Saliency Maps



original image



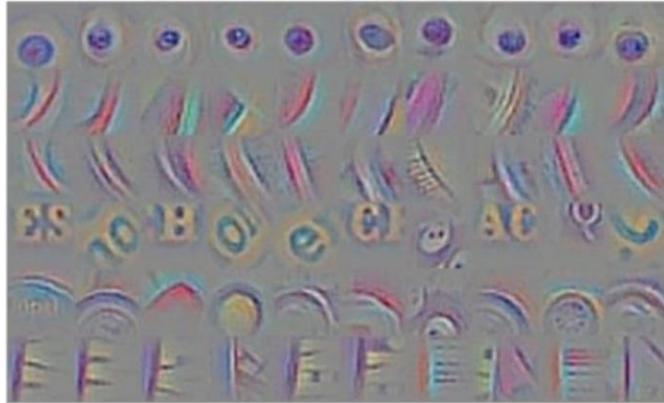
saliency map



map overlay

Guided Backpropagation

guided backpropagation



corresponding image crops



guided backpropagation



corresponding image crops

